Master's Thesis

Riccati-based Primal Regularizations in Interior-Point Solvers for Indefinite Quadratic Programs

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Abstract

In nonlinear model predictive control (NMPC), sequential quadratic programming (SQP) with the exact Hessian of the Lagrangian produces quadratic program (QP) subproblems of which the full Hessian can be indefinite. Structure-exploiting QP solvers embedded in the SQP solver often require a positive definite (PD) full Hessian for carrying out iteration, so the full Hessian is regularized at the SQP level before it is passed to the solver, which in general changes the solution of each QP and thus, in real-time iteration, directly changes the feedback.

This thesis investigates the regularizations that handle the indefiniteness inside interior-point solvers. By regularizing at the QP level, the solver can take advantage of the barrier term in the interior point method (IPM) that adds curvature on the constraints as they become active, without extra estimation of the active set, and the solver still returns the solution of the original QP when the QP is indefinite.

On the algorithmic side, we propose a primal regularization inside the Riccati recursion, namely Riccati-based regularization, which regularizes only at the stages where the Cholesky factorization fails. We show that the minimum perturbation strategy can make the regularization fail, derive an improved strategy, and prove that the regularization becomes inactive near the solution. On the experimental side, for evaluation, we build a benchmark of 175 indefinite optimal control problem (OCP) QPs, collected from the SQP iterations of eight OCPs. We implement the methods in a prototype solver that reproduces the iterations of HPIPM on convex QPs, and compare them with full Hessian regularization and inertia correction in terms of robustness and estimated factorization cost.

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Chapter 1

Introduction

1.1 Motivation

In the field of nonlinear model predictive control (NMPC), in each sampling period, a nonlinear optimal control problem (OCP) is solved to compute the optimal control input for the current state of the system. In order to solve the continuous-time OCPs, direct multiple shooting [1] is a widely used method, which transcribes the OCP into a nonlinear programming problem (NLP) by discretizing the time horizon into multiple shooting intervals [2].

Sequential quadratic programming (SQP) provides a suitable approach for solving the NLP in real-time practice, which linearizes at each iterate to generate a quadratic program (QP) subproblem. In the real-time iteration [3], one SQP iteration is performed per sampling time, warm-started at the shifted previous solution and linearized before the new state is known, so only the QP solve remains between measurement and feedback. The solving of the QP subproblem therefore determines the feedback and the delay, and the structure-exploiting solvers are usually applied for solving with efficiency.

Nevertheless, when the full Hessian of the QP subproblems is provided by the exact Hessian of the Lagrangian [4], it could be indefinite, thus indicating a nonconvex problem. The structure-exploiting solvers embedded in the SQP, such as qpDUNES [5], FORCES [6] and HPIPM [7], however, fail to solve these QPs, for they require convex QPs with a positive definite (PD) full Hessian.

Though FORCES and HPIPM can in principle handle the nonconvex QPs with an indefinite full Hessian as long as the reduced Hessian, i.e. the Hessian on the null space of the dynamics, is PD, the code optimization of both solvers still requires a PD full Hessian, and thus convex QPs are intended by the solvers instead of nonconvex QPs. One method to use these solvers in this situation is to regularize at the SQP level, i.e. to regularize the full Hessian of the QP before it is passed to the solver, which in general modifies the solution of the subproblems and in the real-time iteration the feedback itself.

Given that the augmented reduced Hessian, i.e. the Hessian on the null space of the dynamics and the active constraints, is PD, CONVEXIFY [8] can avoid the modification of the primal solution when the full Hessian is not PD. It applies a backward sweep similar to Riccati recursion to transfer cost between neighboring stages, and thus each stage keeps only the curvature it needs to be PD, so the reduced Hessian remains invariant during this transfer. It also adds curvature by a quadratic penalty on the estimated active constraints with a tunable weight to make the reduced Hessian PD. Together, the full Hessian becomes PD while the augmented reduced Hessian stays unchanged. Thus, the resulting QP can keep the same primal solution if the active set is estimated correctly and the weight is large enough. In the case where the augmented reduced Hessian is indefinite, or the estimation misses some active constraints or the penalty weight is too small, the sweep can still reach a stage with negative curvature after adding the transferred cost and the penalty. At such stages, it falls back to SQP-level regularization by replacing the negative eigenvalues of the stage block with small positive values and continues the sweep, which modifies the QP and thus its solution.

In the interior point method (IPM), however, the barrier term from the inequality constraints enables the addition of these curvatures inherently. Since the curvature contribution increases on the constraints that are becoming active and vanishes on the inactive ones, estimation of the active set and tuning of the penalty weight can be avoided. So when the augmented reduced Hessian is PD, the barrier term can inherently make the reduced Hessian PD without choosing a weight. In other cases, such as in the early iterations, where the augmented reduced Hessian fails to become PD, a regularization to the Hessian is still needed. But if the regularization is made inside the solver, the solution of the original QP remains, and thus an exact Newton step in SQP is preserved, which differs from

the approximate Newton step obtained by SQP-level regularization, and so does the feedback in real-time iteration.

A structure-exploiting interior point QP solver that handles indefiniteness at the QP level is therefore interesting as an alternative to regularizing at the SQP level. And this leaves open how the correction should be made inside the solver. This thesis addresses this question for indefinite QPs with optimal control structure.

1.2 Contributions

The main contributions of this thesis are the following:

1. **Riccati-based regularization**: we propose a regularization inside the Riccati recursion, i.e. Riccati-based regularization, that exploits the OCP structure of the problem, and regularizes the Hessian only when Cholesky factorization fails. We further analyze why the minimum perturbation in this regularization fails, improve the regularization strategy, and prove the regularization to be inactive near the solution (Chapter 3).
2. **A benchmark of** 175 **indefinite OCP QPs**: we provide a pipeline for collecting QPs from the SQP iterations with acados [9], and collect 175 QPs from 8 nonlinear OCPs, together with an analysis of the indefiniteness regarding the benchmark (Chapter 4).
3. **Numerical comparison and ablation**: We implement a prototype solver that shows identical iteration behavior to HPIPM on convex QPs, and perform experiments with this solver on the benchmark to investigate the performance of the methods. In addition, we present an analysis of solving failures and include ablation studies on the solver settings to show which components the results depend on (Chapter 5).

1.3 Outline

The remainder of this thesis is organized as follows. In Chapter 2, we start with theoretical background regarding the primal–dual IPM for OCP-structured QPs, where the Riccati recursion is used to solve the QPs, and introduce the

barrier update and step size control strategies adopted in this work. In Chapter 3 we propose the Riccati-based primal regularization, derive the improved stage regularization strategy from the failure of minimum perturbation, and prove that the regularization is asymptotically inactive. Chapter 4 constructs the benchmark of indefinite QPs from the SQP iterations of eight OCPs, and an analysis is presented about how the indefiniteness is distributed over the stages and where it originates. We carry out experiments in Chapter 5. We first describe the prototype solver and the experiment setup, compare the Riccati-based methods with full Hessian regularization and inertia correction on the benchmark, and discuss the instances on which the methods fail, with ablations for investigating which components the results depend on. Finally, Chapter 6 summarizes the results and outlines directions for future work.

Chapter 2

Primal–dual interior point method for optimal control problems

In this chapter, the theoretical background regarding the IPM is introduced. The primal–dual IPM is introduced first in Section 2.1 as the core algorithm for solving QPs in this thesis. Then, the QP with OCP structure is introduced in Section 2.2, where the Riccati recursion is applied for solving the QP with linear complexity. Finally, the adaptive barrier update and step size control strategies are discussed in Section 2.3 and Section 2.4 with consideration of the nonconvex feature in this work.

2.1 Primal–dual interior point method

Consider a QP with equality and inequality constraints, of which the formulation can be written as:

$$\begin{aligned} \underset{y}{\text{minimize}} \quad & \frac{1}{2} y^\top H y + g^\top y \\ \text{subject to} \quad & Fy = b, \\ & Gy \leq d, \end{aligned} \tag{2.1}$$

where y are the decision variables, H represents the full Hessian, g is the linear term, and F and G stand for the linear constraint matrices for equality and inequality. Since the constraints are affine w.r.t. y, the definiteness of the full Hessian matrix H determines whether the QP in Equation (2.1) is convex or nonconvex.

By introducing slack variables t, we have the following formulation:

$$\begin{aligned} \underset{y,\,t}{\text{minimize}} \quad & \frac{1}{2} y^\top H y + g^\top y \\ \text{subject to} \quad & Fy = b, \\ & Gy + t = d, \\ & t \geq 0. \end{aligned} \tag{2.2}$$

We can further obtain the Lagrangian function of the QP in Equation (2.2) as

$$\mathcal{L}(y, t, \pi, \lambda) = \frac{1}{2} y^\top H y + g^\top y + \pi^\top (Fy - b) + \lambda^\top (Gy + t - d), \tag{2.3}$$

where π and $\lambda \geq 0$ are the Lagrange multipliers associated with the equality and inequality constraints, respectively. Then the Karush–Kuhn–Tucker (KKT) conditions of the QP in Equation (2.2) are given by

$$Hy + g + F^\top \pi + G^\top \lambda = 0, \tag{2.4a}$$
$$Fy - b = 0, \tag{2.4b}$$
$$Gy + t - d = 0, \tag{2.4c}$$
$$\lambda_j \, t_j = 0, \quad j = 1, \ldots, n_c, \tag{2.4d}$$
$$\lambda \geq 0, \quad t \geq 0, \tag{2.4e}$$

where n_c is the number of inequality constraints. In the primal–dual IPM, a homotopy parameter τ, also called the barrier parameter, is applied to Equation (2.4d), which is the only nonlinear equation, for relaxing the complementarity. Equation (2.4d) can be rewritten as:

$$\lambda_j \, t_j = \tau, \quad j = 1, \ldots, n_c, \tag{2.5}$$

and the parameter τ is shrunk towards zero with a certain strategy during the iterations to obtain the solution of the original problem, and step size control

is applied to ensure the satisfaction of the inequalities in Equation (2.4e). The corresponding update strategy and step size control will be discussed in the following sections.

In order to solve the nonlinear system in Equation (2.4), Newton's method linearizes the system at each iteration i to obtain the step $(\Delta y, \Delta\pi, \Delta\lambda, \Delta t)$ from the linear system, which takes the form of:

$$\begin{bmatrix} H & F^\top & G^\top & 0 \\ F & 0 & 0 & 0 \\ G & 0 & 0 & I \\ 0 & 0 & T_i & \Lambda_i \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta\pi \\ \Delta\lambda \\ \Delta t \end{bmatrix} = - \begin{bmatrix} r^i_{\text{stat}} \\ r^i_{\text{eq}} \\ r^i_{\text{ineq}} \\ r^i_{\text{comp}} \end{bmatrix}, \tag{2.6}$$

where the diagonal matrices $T_i = \text{diag}(t_i)$, $\Lambda_i = \text{diag}(\lambda_i)$, and the perturbed KKT residuals follow as:

$$\begin{bmatrix} r^i_{\text{stat}} \\ r^i_{\text{eq}} \\ r^i_{\text{ineq}} \\ r^i_{\text{comp}} \end{bmatrix} := \begin{bmatrix} Hy_i + g + F^\top\pi_i + G^\top\lambda_i \\ Fy_i - b \\ Gy_i + t_i - d \\ T_i\Lambda_i\mathbf{1} - \tau_i\mathbf{1} \end{bmatrix}. \tag{2.7}$$

Based on the structure of the KKT system in Equation (2.6), eliminating Δt and $\Delta\lambda$ from the KKT system in Equation (2.6) by a Schur complement yields the symmetric saddle point system:

$$\begin{bmatrix} \tilde{H}_i & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta\pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}^i_{\text{stat}} \\ r^i_{\text{eq}} \end{bmatrix}, \tag{2.8}$$

with $\tilde{H}_i = H + G^\top T_i^{-1}\Lambda_i G \succeq H$ and $\tilde{r}^i_{\text{stat}} = r^i_{\text{stat}} + G^\top T_i^{-1}(\Lambda_i r^i_{\text{ineq}} - r^i_{\text{comp}})$. Notably, the barrier term $G^\top T_i^{-1}\Lambda_i G$ can be regarded as the curvature contribution from the inequality constraints. The weight on each constraint, i.e. the corresponding diagonal entry of $T_i^{-1}\Lambda_i$, increases as the constraint becomes active and tends to zero if it is inactive, and this barrier term is positive semidefinite, since t and λ are kept strictly positive throughout the iterations.

In the reduced KKT system of Equation (2.8), assuming F has full row rank, the

null space method can be applied to further reduce the system. Let the columns of Z form a basis of the null space of F, i.e. $FZ = 0$, and the columns of Y span the range of $F^\top$, so that $[\,Y \;\; Z\,]$ is nonsingular and FY is invertible.

The primal step can then be decomposed uniquely as $\Delta y = Y\,\Delta y_y + Z\,\Delta y_z$. Since $F\Delta y = FY\,\Delta y_y$, only the range-space component enters the equality constraints. This yields a system exposing the reduced Hessian as follows:

$$\tilde{M}_i\,\Delta y_z = -\Big(Z^\top \tilde{r}^i_{\text{stat}} + Z^\top \tilde{H}_i\,Y\,\Delta y_y\Big), \tag{2.9}$$

where $Z^\top$ eliminates the equality Jacobian with $Z^\top F^\top = 0$, feasibility fixes the range-space component $\Delta y_y = -(FY)^{-1} r^i_{\text{eq}}$, and $\tilde{M}_i = Z^\top(H + G^\top T_i^{-1}\Lambda_i G)Z = Z^\top \tilde{H}_i Z$ is the reduced Hessian including the curvature contribution.

2.2 Efficient solution with Riccati recursion

In the context of optimal control, the decision variables y can be partitioned into state and control variables, and the QP in Equation (2.1) takes the particular form:

$$\begin{aligned}
\underset{x,\,u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \left(\frac{1}{2} \begin{bmatrix} x_k \\ u_k \end{bmatrix}^\top \begin{bmatrix} Q_k & S_k^\top \\ S_k & R_k \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix} + \begin{bmatrix} g_k^x \\ g_k^u \end{bmatrix}^\top \begin{bmatrix} x_k \\ u_k \end{bmatrix} \right) \\
& + \frac{1}{2} x_N^\top Q_N x_N + (g_N^x)^\top x_N \\
\text{subject to} \quad & x_0 = \hat{x}_0, \\
& x_{k+1} = A_k x_k + B_k u_k + b_k, \quad k = 0, \ldots, N-1, \\
& G_k^x x_k + G_k^u u_k \le d_k, \qquad\quad k = 0, \ldots, N-1, \\
& G_N^x x_N \le d_N,
\end{aligned} \tag{2.10}$$

where $x_k \in \mathbb{R}^{n_x}$ and $u_k \in \mathbb{R}^{n_u}$ are the state and control at stage k, and N is the horizon length. $Q_k \in \mathbb{R}^{n_x \times n_x}$, $R_k \in \mathbb{R}^{n_u \times n_u}$, and $S_k \in \mathbb{R}^{n_u \times n_x}$ are the state, control, and state–control coupling blocks of the stage Hessian H_k, $g_k^x \in \mathbb{R}^{n_x}$ and $g_k^u \in \mathbb{R}^{n_u}$ are the corresponding gradients, and Q_N, g_N^x are the terminal Hessian and gradient. $A_k \in \mathbb{R}^{n_x \times n_x}$ and $B_k \in \mathbb{R}^{n_x \times n_u}$ are the Jacobians of the discretized dynamics with respect to the state and control, $b_k \in \mathbb{R}^{n_x}$ is the dynamics residual, and $\hat{x}_0 \in \mathbb{R}^{n_x}$ is the initial state. $G_k^x \in \mathbb{R}^{n_k^c \times n_x}$ and $G_k^u \in \mathbb{R}^{n_k^c \times n_u}$ are the Jacobians

of the n_k^c inequality constraints at stage k, including linear constraints and box constraints on states and controls, with right-hand side $d_k \in \mathbb{R}^{n_k^c}$.

Following the same derivation as in Section 2.1, we can obtain the KKT system of Equation (2.10), which exhibits a tridiagonal structure since the cost and inequality constraints are stage-wise and the dynamics couple only consecutive stages:

$$\begin{bmatrix} \ddots & & & & & \\ & 0 & -I & & & \\ & -I & \begin{pmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{pmatrix} & \begin{matrix} A_k^\top \\ B_k^\top \end{matrix} & \\ & 0 & A_k \quad B_k & 0 & -I \\ & & & -I & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ \Delta\pi_k \\ \Delta x_k \\ \Delta u_k \\ \Delta\pi_{k+1} \\ \vdots \end{bmatrix} = - \begin{bmatrix} \vdots \\ r_{\text{eq}}^{k-1} \\ \tilde{r}_{\text{stat},x}^k \\ \tilde{r}_{\text{stat},u}^k \\ r_{\text{eq}}^k \\ \vdots \end{bmatrix}, \tag{2.11}$$

where the block matrices $\tilde{Q}_k$, $\tilde{R}_k$, and $\tilde{S}_k$ contain the barrier term as:

$$\begin{aligned} \tilde{Q}_k &= Q_k + (G_k^x)^\top \Lambda_k T_k^{-1} G_k^x, \\ \tilde{R}_k &= R_k + (G_k^u)^\top \Lambda_k T_k^{-1} G_k^u, \\ \tilde{S}_k &= S_k + (G_k^u)^\top \Lambda_k T_k^{-1} G_k^x. \end{aligned}$$

This tridiagonal structure allows for efficient solution of the KKT system using Riccati recursion [10], which consists of a backward and forward pass. The backward pass computes the feedback gains K and cost-to-go matrices P, while the forward pass computes the optimal state and control trajectories. The algorithmic details of the Riccati recursion can be summarized in Algorithm 1.

By applying the Riccati recursion, we solve the KKT system in Equation (2.9) without computing the inverse of the reduced Hessian $\tilde{M}_i$, and the computational complexity is reduced from $\mathcal{O}(N^3)$ for a dense factorization to $\mathcal{O}(N)$ [10], thus achieving an efficient solution by exploiting the OCP structure.

2.3 Adaptive barrier update

In this section, two closely related adaptive barrier update schemes are discussed for the primal–dual IPM. While the monotone decrease of the barrier parameter τ,

Algorithm 1 Riccati Recursion for Solving OCP Structure QP

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\text{stat},x}$

// Backward recursion
for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$
  $K_k \leftarrow -\tilde{M}_k^{-1} \Gamma_k$
  $k_k \leftarrow -\tilde{M}_k^{-1} \left(\tilde{r}^k_{\text{stat},u} + B_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) \right)$
  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k$
  $p_k \leftarrow \tilde{r}^k_{\text{stat},x} + A_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

// Forward recursion
for $k = 0$ **to** $N-1$ **do**
  $\Delta u_k \leftarrow K_k \Delta x_k + k_k$
  $\Delta x_{k+1} \leftarrow A_k \Delta x_k + B_k \Delta u_k + r^k_{\text{eq}}$
  $\Delta \pi_{k+1} \leftarrow P_{k+1} \Delta x_{k+1} + p_{k+1}$
  $\Delta t_k \leftarrow -(G^x_k \Delta x_k + G^u_k \Delta u_k + r^k_{\text{ineq}})$
  $\Delta \lambda_k \leftarrow T_k^{-1}(-r^k_{\text{comp}} - \Lambda_k \Delta t_k)$
end for

such as the Fiacco–McCormick scheme [11], enjoys strong convergence guarantees, it may lead to slow convergence and practical limitations [12]. We therefore prefer a more aggressive and adaptive update of τ to achieve faster convergence in optimal control.

2.3.1 Mehrotra's predictor–corrector

For convex QPs, Mehrotra's predictor–corrector scheme [13] is usually considered a good choice for fast convergence in optimal control, which adaptively updates the barrier parameter τ and adds a second-order corrector for improving the direction. In the scheme, the barrier parameter τ is defined as:

$$\tau_i = \sigma_i \mu_i, \qquad \mu_i = \frac{t_i^\top \lambda_i}{n_c}, \tag{2.12}$$

where μ_i is the duality measure, and $\sigma_i \geq 0$ is the centering parameter that controls the trade-off between centering and aggressive steps. In each iteration,

two solves of Equation (2.6) that share the same left-hand side (LHS) and differ only in the complementarity at the right-hand side (RHS) are performed, and each iteration consists of a predictor (affine scaling) step, a centering update, and a corrector step.

Predictor step. The scheme starts with computing a predictor step, also called the affine scaling step, for which the linear system in Equation (2.6) is solved with $\sigma_i = 0$ in the complementarity residual r^i_{comp} and no correction considered, i.e. the resulting complementarity residual is written as:

$$r^i_{\text{comp,aff}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1} = T_i \Lambda_i \mathbf{1}, \tag{2.13}$$

to obtain the affine scaling direction $(\Delta y_{\text{aff}}, \Delta \pi_{\text{aff}}, \Delta \lambda_{\text{aff}}, \Delta t_{\text{aff}})$. The affine step is used to estimate the centering parameter σ_i and form the second-order correction in the following.

Centering update. After the predictor step is obtained, the centering parameter σ is updated based on the estimated duality measure μ_{aff} after the affine step is taken, which is defined as:

$$\sigma_i = (\mu_{\text{aff}}/\mu_i)^3 \tag{2.14}$$

$$\mu_{\text{aff}} = \frac{(t_i + \alpha^i_{\text{prim,aff}} \Delta t_{\text{aff}})^\top (\lambda_i + \alpha^i_{\text{dual,aff}} \Delta \lambda_{\text{aff}})}{n_c}, \tag{2.15}$$

where the step sizes $\alpha^i_{\text{prim,aff}}$ and $\alpha^i_{\text{dual,aff}}$ are the largest step sizes that keep the slack variables and Lagrange multipliers nonnegative, i.e.

$$\alpha^i_{\text{prim,aff}} = \min \left(1, \min_{j:\Delta t_{\text{aff},j}<0} -\frac{t^i_j}{\Delta t_{\text{aff},j}} \right), \tag{2.16}$$

$$\alpha^i_{\text{dual,aff}} = \min \left(1, \min_{j:\Delta \lambda_{\text{aff},j}<0} -\frac{\lambda^i_j}{\Delta \lambda_{\text{aff},j}} \right). \tag{2.17}$$

If $\mu_{\text{aff}} \ll \mu_i$, the affine step achieves a substantial reduction in the duality gap, which sharply gives $\sigma_i \to 0$ (near-Newton, aggressive), while a stalled step ($\mu_{\text{aff}} \approx \mu_i$) gives $\sigma_i \to 1$ (recentering towards the central path).

Corrector step. In the corrector step, the linear system in Equation (2.6) is solved again where the complementarity residual is modified to include the updated centering parameter σ_i and a second-order correction term using the calculated affine steps as follows:

$$r^i_{\text{comp,cor}} = T_i\Lambda_i\mathbf{1} + \Delta T_{\text{aff}}\Delta\Lambda_{\text{aff}}\mathbf{1} - \sigma_i\mu_i\mathbf{1}, \tag{2.18}$$

where $\Delta T_{\text{aff}} = \text{diag}(\Delta t_{\text{aff}})$, $\Delta\Lambda_{\text{aff}} = \text{diag}(\Delta\lambda_{\text{aff}})$, the term $\Delta T_{\text{aff}}\Delta\Lambda_{\text{aff}}\mathbf{1}$ cancels the second-order error that the affine step leaves in $t_j\lambda_j$, and $-\sigma_i\mu_i\mathbf{1}$ supplies the centering. This second solve returns the final direction $(\Delta y, \Delta\pi, \Delta\lambda, \Delta t)$ directly, and a step size control will follow to ensure the positivity of the slack variables and Lagrange multipliers.

Condition on Corrector Step. In practice, the corrector step is usually applied conditionally. The mechanism is, therefore, called conditional Mehrotra's predictor–corrector scheme [12], where the corrector step is only accepted if it does not largely increase the duality measure after the corrector step is taken compared with that after the predictor step, i.e.,

$$\mu_{\text{cor}} \leq 2\mu_{\text{aff}}, \tag{2.19}$$

$$\mu_{\text{cor}} = \frac{(t_i + \alpha^i_{\text{prim,cor}}\Delta t)^\top(\lambda_i + \alpha^i_{\text{dual,cor}}\Delta\lambda)}{n_c} \tag{2.20}$$

where $\alpha^i_{\text{prim,cor}}$ and $\alpha^i_{\text{dual,cor}}$ are computed as the affine step sizes in Equation (2.16), but along the corrected direction. Otherwise, Equation (2.6) is solved again without the correction term.

2.3.2 Mehrotra's probing

Different from Mehrotra's predictor–corrector or conditional Mehrotra, Mehrotra's probing scheme [12] drops the corrector step entirely, resulting in a more conservative approach for updating the barrier parameter τ by only considering the predictor step and centering update. This gives the so-called centering step,

and the corresponding complementarity residual at the RHS is given by:

$$r^i_{\text{comp,pro}} = T_i \Lambda_i \mathbf{1} - \sigma_i \mu_i \mathbf{1}, \tag{2.21}$$

where the centering parameter σ_i is updated using Equation (2.14) after the affine step.

Comparing the two schemes, we would argue that Mehrotra's predictor–corrector scheme is not necessarily the best choice in the nonconvex case. In the corrector step, $\Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1}$ is added to the complementarity residual to include the complementarity error of a full affine step, which assumes the affine direction is an accurate Newton step. In nonconvex QPs, however, the Hessian at the LHS usually needs to be modified, and thus the direction can be adjusted for descent when the constraints are satisfied, so it is actually an inexact Newton step, violating the assumption of the predictor–corrector scheme.

Probing uses the affine step only through $\sigma = (\mu_{\text{aff}}/\mu)^3$, so an inexact affine step just gives $\sigma \approx 1$, i.e. a centering step. And the iterate moves back to the central path instead of trying to decrease the incorrect complementarity error brought by the inexact Newton step.

2.4 Step size control

In this section, we discuss the step size control employed in the primal–dual IPM. The fraction-to-the-boundary (FTB) rule is first introduced as a common strategy for step size control in the IPM, which ensures that the updated slack variables and Lagrange multipliers remain positive. Then, a two-phase line search with backtracking for QPs is introduced, which is attempted for the nonconvex case.

2.4.1 Fraction-to-the-boundary rule

The FTB rule is a common strategy for step size control in the primal–dual IPM. The maximal step sizes $\alpha^i_{\text{prim,max}}$ and $\alpha^i_{\text{dual,max}}$ are computed without violating

the nonnegativity of the slack variables and Lagrange multipliers, as:

$$\alpha^i_{\text{prim,max}} = \min_{j:\Delta t_j<0} -\frac{t^i_j}{\Delta t_j}, \tag{2.22}$$

$$\alpha^i_{\text{dual,max}} = \min_{j:\Delta \lambda_j<0} -\frac{\lambda^i_j}{\Delta \lambda_j}. \tag{2.23}$$

Then, the actual step sizes are chosen as a fraction of the maximum step sizes, i.e.,

$$\alpha^i_{\text{prim}} = \min(1, \eta\alpha^i_{\text{prim,max}}), \tag{2.24}$$

$$\alpha^i_{\text{dual}} = \min(1, \eta\alpha^i_{\text{dual,max}}), \tag{2.25}$$

where $\eta \in (0, 1)$ is a parameter that controls the fraction of the maximum step size to be taken, typically set to a value close to 1 to ensure sufficient progress while maintaining feasibility.

2.4.2 Two-phase line search

In nonconvex cases, the effect of line search still can be investigated. We thus apply a two-phase line search inspired by the funnel method [14] after the FTB part is performed.

In the IPM, the objective ϕ used in the line search is the barrier objective

$$\phi(y, t) = \frac{1}{2}y^\top Hy + g^\top y - \tau \sum_{j=1}^{n_c} \ln t_j, \tag{2.26}$$

which is minimized subject to $Fy = b$ and $Gy + t = d$. With $\lambda_j = \tau/t_j$, the corresponding KKT conditions recover the relaxed complementarity in Equation (2.5). The line search consists of three parts: a switching condition, the f-phase and the h-phase.

The *switching condition* decides which phase the trial step belongs to. It depends only on the current iterate and the trial step size, so it is evaluated before a candidate is formed.

$$-\alpha^i_{\text{prim}}\rho_i > \kappa_{\text{s}}\,\theta_i^2, \qquad \kappa_{\text{s}} > 0, \tag{2.27}$$

where $\theta_i = \|(r^i_{\text{eq}}, r^i_{\text{ineq}})\|$ is the constraint violation at the current iterate and $\rho_i = \nabla\phi_i^\top \Delta z$ the directional derivative of ϕ along the primal search direction $\Delta z = (\Delta y, \Delta t)$.

If Equation (2.27) fails, the predicted objective decrease is small relative to the constraint violation, and the iterate is in the *h-phase*. The FTB step is then accepted without evaluating the objective. The constraints of the QP are affine and the Newton direction solves the corresponding rows of the linear system exactly, so $\theta(\alpha^i_{\text{prim}}) = (1 - \alpha^i_{\text{prim}})\,\theta_i$ and the largest admissible step reduces the constraint violation the most.

If Equation (2.27) holds, sufficient objective progress is expected and the iterate is in the *f-phase*. A candidate iterate $z^+(\alpha^i_{\text{prim}})$ is formed in the primal variables (y, t) and accepted if its objective ϕ^+ satisfies the Armijo condition [15]

$$\phi^+ \leq \phi_i + \gamma\,\alpha^i_{\text{prim}}\,\rho_i, \qquad \gamma \in (0, 1). \tag{2.28}$$

Otherwise, $\alpha^i_{\text{prim}} \leftarrow \beta\,\alpha^i_{\text{prim}}$ with $\beta \in (0, 1)$ and the search repeats. If all $n_{\max}$ trials are rejected, the FTB step size is restored. The dual step size α^i_{dual} is not changed. The procedure is summarized in Algorithm 2.

Algorithm 2 Two-phase line search

Input: iterate z_i, direction Δz, FTB steps $\alpha^i_{\text{prim}}, \alpha^i_{\text{dual}}$, ϕ_i, θ_i, ρ_i
$\alpha^i_{\text{prim,FTB}} \leftarrow \alpha^i_{\text{prim}}$
for ls $= 1$ **to** $n_{\max}$ **do**
  if $\alpha^i_{\text{prim}}(-\rho_i) \leq \kappa_{\text{s}}\,\theta_i^2$ **then** ▷ h-phase: feasibility dominates
    break ▷ accept FTB-scaled step
  end if
  form candidate $z^+(\alpha^i_{\text{prim}})$
  compute ϕ^+
  if $\phi^+ \leq \phi_i + \gamma\,\alpha^i_{\text{prim}}\,\rho_i$ **then** ▷ f-phase: Armijo
    break
  end if
  if ls $= n_{\max}$ **then**
    $\alpha^i_{\text{prim}} \leftarrow \alpha^i_{\text{prim,FTB}}$ ▷ all trials rejected: revert to the FTB step
  else
    $\alpha^i_{\text{prim}} \leftarrow \beta\,\alpha^i_{\text{prim}}$
  end if
end for
return $\alpha^i_{\text{prim}},\ \alpha^i_{\text{dual}}$

Two simplifications distinguish this procedure from the funnel method of [14]. First, no funnel bound $\theta_{\max}$ is maintained. In the funnel method, a trial iterate is admissible only if $\theta(z) \leq \theta_{\max}$, and the bound $\theta_{\max}$ is decreased whenever an h-step is accepted. Here $\theta(\alpha^i_{\text{prim}}) = (1 - \alpha^i_{\text{prim}})\,\theta_i \leq \theta_i$ for every $\alpha^i_{\text{prim}} \in (0, 1]$, so no trial step increases the constraint violation, and the funnel itself can then be removed. Second, since H is generally indefinite, $\Delta y^\top H \Delta y$ can have either sign, and the second-order term is unreliable for classifying the step. Instead of considering the second-order model as in [14], the switching condition in Equation (2.27) uses only the first-order term $-\rho_i$, as the Armijo condition in Equation (2.28) does. To compensate, κ_{s} is set small, so that objective-reducing steps are not classified as h-phase steps.

Chapter 3

Primal regularization for indefinite quadratic programs

In this chapter, we focus on QPs with indefinite full Hessian matrices, referred to as indefinite QPs. We first review the numerical strategies to compute the adjustment matrix, and motivate the reduced Hessian as the target for the detection and the regularization in Section 3.1. In Section 3.2, we then propose the Riccati-based primal regularization, which detects the indefiniteness and applies the regularization inside the recursion. Next, we analyze in Section 3.3 why the minimum perturbation rule fails in this setting, and derive an improved stage regularization strategy from that analysis. Finally, in Section 3.4, we prove that, owing to the detection on the reduced Hessian, the regularization becomes asymptotically inactive as the iterates approach the solution.

3.1 Primal regularization

In indefinite QPs, the regularization needs to be applied to the Hessian matrix to ensure a descent search direction in the IPM when the constraints are satisfied. The resulting regularized reduced KKT system can be generally presented as:

$$\begin{bmatrix} \tilde{H}_i + E_i & F^\top \\ F & 0 \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta \pi \end{bmatrix} = - \begin{bmatrix} \tilde{r}^i_{\text{stat}} \\ r^i_{\text{eq}} \end{bmatrix}, \tag{3.1}$$

where E_i is the adjustment matrix that renders the Hessian PD at the i-th iteration. This process is referred to as primal regularization [16].

We start with the numerical strategies to compute E_i in Section 3.1.1, then discuss the detection of the indefiniteness in Section 3.1.2, and finally the regularization on the full or the reduced Hessian in Section 3.1.3.

3.1.1 Regularization strategies

In order to construct the adjustment matrix E_i, three categories of strategies are commonly used, which are eigenvalue modification (EM), modified Cholesky factorization (MC) and Levenberg–Marquardt (LM) [17–19]. From these categories we select four methods to be implemented and compared: PROJECT and MIRROR from EM [8], GMW from MC [20], and inertia correction from LM [16].

Assume a matrix A admits the eigenvalue decomposition $A = VD_EV^\top$ and the modified Cholesky factorization $A = LD_IL^\top$, where V is an orthogonal matrix of eigenvectors, L is a unit lower triangular matrix, D_E is a diagonal matrix of real eigenvalues, and D_I is a diagonal matrix whose signs give the inertia of A. In this notation, the four methods read as follows:

PROJECT. This approach clips any negative or near-zero eigenvalues to a small positive constant $\epsilon > 0$, which keeps a minimal adjustment to the original matrix for positive definiteness, thus achieving the minimal Frobenius norm of the adjustment matrix E [17], but may lead to an increase in the condition number if the original eigenvalues were not close to zero. The operation can be described as:

$$\text{project}(A, \epsilon) := V \max(\epsilon I, D_E)V^\top,$$

where the max operator is applied component-wise.

MIRROR. This method reflects negative eigenvalues to their positive counterparts. It preserves the magnitude of the curvature in all directions, which avoids the ill-conditioning of PROJECT at the cost of a larger perturbation when

$|\nu_{\min}| > \epsilon$. The operation can be described as:

$$\text{mirror}(A, \epsilon) := V \max(\epsilon I, |D_E|) V^\top.$$

GMW. Unlike EM, which corrects the eigenvalues, GMW corrects the pivots of the $\text{LDL}^\top$ factorization, whose signs give the inertia of A [20]. During the GMW factorization, each pivot is replaced by the largest of three values: its absolute value which flips the value if negative and remains the same if positive, a small value ϵ for lifting a near-zero pivot, and the smallest value for keeping the entries of the corresponding column of $LD_I^{1/2}$ below a fixed bound.

The increments form a nonnegative diagonal correction E with $A + E = LD_I L^\top$, which avoids the cost of an eigendecomposition and still guarantees positive definiteness, but E is generally not minimal and tends to over-modify A.

Inertia Correction. In LM, all decompositions or factorizations are abandoned. A uniform multiple of the identity is instead added to shift every eigenvalue by the same amount. The shift avoids the cost of any extra operation and is easy to implement. However, because it acts equally in all directions, a shift large enough for a few negative eigenvalues also distorts the eigenvalues that were already positive. The operation can be described as:

$$\text{ic}(A, \Delta) := A + \Delta I, \qquad \Delta \geq 0.$$

In practice, Δ needs to be found by heuristics or by repeatedly increasing Δ until every diagonal entry of D_I is positive.

3.1.2 Indefiniteness detection on reduced Hessian

Though the positive definiteness of the full Hessian H is rather easy to detect and is sufficient for convexity, an equality-constrained QP only requires the reduced Hessian to be PD for a unique global minimum. The following theorem, stated in [8] (for a proof, see [17]), makes this precise:

Theorem 1. *Assume that linear independence constraint qualification (LICQ) holds. A QP with only equality constraints has a unique global optimum if and only if the reduced Hessian is PD:*

$$M = Z^\top H Z \succ 0. \tag{3.2}$$

The positive definiteness of the reduced Hessian is thus a necessary and sufficient condition for the equality-constrained QP admitting a unique global optimum.

Since the KKT system after the Schur complement in Equation (2.8) is that of an equality-constrained QP as:

$$\begin{aligned} \underset{\Delta y}{\text{minimize}} \quad & \frac{1}{2}\Delta y^\top \tilde{H}_i \Delta y + (\tilde{r}^i_{\text{stat}})^\top \Delta y \\ \text{subject to} \quad & F\Delta y + r^i_{\text{eq}} = 0, \end{aligned} \tag{3.3}$$

by Theorem 1, the step Δy is the unique minimizer of this QP if and only if $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$.

Based on this, a detection of the indefiniteness of $\tilde{M}_i$ is therefore preferable to that of $\tilde{H}_i$, which avoids false alarms when $\tilde{H}_i$ is indefinite but $\tilde{M}_i \succ 0$, and hence unnecessary regularization.

3.1.3 Regularization on full or reduced Hessian

Once the indefiniteness is detected, a correction should be applied to the Hessian so that Δy can be a descent direction at the feasible iterates. Either the full or the reduced Hessian can be the target matrix, which the correction can be derived from and applied to.

The full Hessian regularization takes the full Hessian $\tilde{H}_i$ as the target. In optimal control, it is applied stage-wise to the block $\tilde{H}_k$ of each stage k to preserve the sparse structure,

$$\tilde{H}^{\text{reg}}_k = \begin{bmatrix} \tilde{Q}_k & \tilde{S}_k^\top \\ \tilde{S}_k & \tilde{R}_k \end{bmatrix} + E_k \succ 0, \quad k = 0, 1, \ldots, N-1. \tag{3.4}$$

This is simple to implement, each E_k depends only on $\tilde{H}_k$, requiring neither the constraints nor a null-space basis, and it guarantees a descent direction at infeasible iterates, i.e. $\Delta y^\top \tilde{H}_i^{\text{reg}} \Delta y > 0$ for $\Delta y = Y\Delta y_y + Z\Delta y_z$ with $\Delta y_y \neq 0$. However, it could be conservative when equality constraints are satisfied. At an iterate with $r_{\text{eq}}^i = 0$, Δy lies in the null space of F, so only the curvature of $\tilde{H}_i$ within this subspace affects the step, and removing negative curvature outside it makes the adjustment larger than necessary.

With a part adapted from [17, Theorem 16.6], the following result shows that, when equality constraints are satisfied, positive curvature in the null space is both necessary and sufficient for a descent direction.

Theorem 2. *Assume that LICQ holds, $r_{\text{eq}}^i = 0$, and $\tilde{M}_i = Z^\top \tilde{H}_i Z$ is nonsingular. Then $(\tilde{r}_{\text{stat}}^i)^\top \Delta y < 0$ holds for every $\tilde{r}_{\text{stat}}^i$ with $Z^\top \tilde{r}_{\text{stat}}^i \neq 0$ if and only if $\tilde{M}_i \succ 0$.*

Proof. Since $r_{\text{eq}}^i = 0$, the range-space component $\Delta y_y = -(FY)^{-1} r_{\text{eq}}^i$ vanishes, so $\Delta y = Z\Delta y_z$ and Equation (2.9) reduces to $\tilde{M}_i \Delta y_z = -Z^\top \tilde{r}_{\text{stat}}^i$. Hence

$$(\tilde{r}_{\text{stat}}^i)^\top \Delta y = -(Z^\top \tilde{r}_{\text{stat}}^i)^\top \tilde{M}_i^{-1} (Z^\top \tilde{r}_{\text{stat}}^i). \tag{3.5}$$

For sufficiency, if $\tilde{M}_i \succ 0$, then $\tilde{M}_i^{-1} \succ 0$, and the RHS of Equation (3.5) is negative for every $Z^\top \tilde{r}_{\text{stat}}^i \neq 0$.

For necessity, we proceed by contraposition. Suppose that $\tilde{M}_i$ is not PD. Being nonsingular, it has an eigenpair (ν, v) with $\nu < 0$. Since Z has full column rank, $\tilde{r}_{\text{stat}}^i = Z(Z^\top Z)^{-1} v$ satisfies $Z^\top \tilde{r}_{\text{stat}}^i = v \neq 0$, and Equation (3.5) with $\tilde{M}_i^{-1} v = v/\nu$ gives $(\tilde{r}_{\text{stat}}^i)^\top \Delta y = -v^\top v/\nu > 0$, so Δy is an ascent direction for this residual. □

At feasible iterates, a PD reduced Hessian is therefore exactly what a descent direction requires, and the reduced Hessian regularization, which takes $\tilde{M}_i$ as the target, enforces this condition.

However, $\tilde{M}_i$ is usually not directly available. Computing it requires a null-space basis of the equality constraints, e.g. via condensing in optimal control [21], which not only introduces extra computational cost but also yields a dense $\tilde{M}_i$, losing the

stage-wise structure preserved by the full Hessian regularization. When both the detection and the regularization operate on this dense matrix, the factorization cost grows cubically with the horizon N instead of linearly as in the Riccati recursion, which contradicts the structure-exploiting purpose.

3.2 Riccati-based primal regularization

We thus propose a Riccati-based primal regularization method, which naturally obtains the reduced Hessian $\tilde{M}_i$ implicitly via Riccati recursion. The Riccati recursion serves two purposes. First, it detects the indefiniteness of the reduced Hessian $\tilde{M}_i$ with Cholesky factorization as the sign. Second, it regularizes the reduced Hessian $\tilde{M}_i$ stage-wise to ensure a PD reduced Hessian $\tilde{M}_i$ while preserving the sparse structure of the OCP.

3.2.1 Block reduced Hessian from Riccati recursion

In this subsection, we will restate and establish the connection between the Riccati recursion and the block $\mathrm{LDL}^\top$ factorization of the reduced Hessian $\tilde{M}_i$, which provides not only a way to compute the reduced Hessian $\tilde{M}_i$ implicitly, but also a way to detect the indefiniteness of $\tilde{M}_i$.

Algorithm 3 Backward Pass of Riccati Recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\mathrm{stat},x}$

for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $L_k \leftarrow \mathrm{Cholesky}(\tilde{M}_k)$
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}^k_{\mathrm{stat},u} + B_k^\top (P_{k+1} r^k_{\mathrm{eq}} + p_{k+1})\right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}^k_{\mathrm{stat},x} + A_k^\top (P_{k+1} r^k_{\mathrm{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

From the perspective of the Riccati recursion, the backward pass proceeds in practice as shown in Algorithm 3. Compared with Algorithm 1, the inverse $\tilde{M}_k^{-1}$ is not formed, but $\tilde{M}_k$ is factorized by Cholesky factorization and K_k, k_k follow from triangular solves, which is more efficient and succeeds only if $\tilde{M}_k \succ 0$.

From the perspective of the block $\mathrm{LDL}^\top$ factorization, recall that the second-order term of Equation (3.3) is stage-wise in optimal control:

$$\frac{1}{2}\sum_{k=0}^{N-1}\begin{bmatrix}\Delta x_k\\ \Delta u_k\end{bmatrix}^\top\begin{bmatrix}\tilde{Q}_k & \tilde{S}_k^\top\\ \tilde{S}_k & \tilde{R}_k\end{bmatrix}\begin{bmatrix}\Delta x_k\\ \Delta u_k\end{bmatrix}+\frac{1}{2}\Delta x_N^\top\tilde{Q}_N\Delta x_N.$$

Condensing eliminates the states through the dynamics $\Delta x_{k+1} = A_k\Delta x_k + B_k\Delta u_k + r_{\mathrm{eq}}^k$ and permutes the dense reduced Hessian $\tilde{M}_i$ in $\Delta u_0, \ldots, \Delta u_{N-1}$. Since Δu_{N-1} enters only the last stage term and Δx_N, its diagonal block is

$$(\tilde{M}_i)_{N-1,N-1} = \tilde{R}_{N-1} + B_{N-1}^\top\tilde{Q}_N B_{N-1},$$

which is exactly $\tilde{M}_{N-1}$ in Algorithm 3 with $P_N = \tilde{Q}_N$.

After eliminating Δu_{N-1}, the Schur complement is the reduced Hessian of the problem on stages $0, \ldots, N-2$ with terminal weight P_{N-1}, which has the same structure. Repeating this backward in time to Δu_0 gives the pivots $\tilde{M}_k$ and the block $\mathrm{LDL}^\top$ factorization

$$\tilde{M}_i = L\begin{bmatrix}\tilde{M}_{N-1} & 0 & \cdots & 0\\ 0 & \tilde{M}_{N-2} & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots\\ 0 & 0 & \cdots & \tilde{M}_0\end{bmatrix}L^\top$$

where L is unit lower triangular, with off-diagonal blocks built from the feedback gains K_k and the dynamics.

This confirms that the matrices $\tilde{M}_k$ from the Riccati recursion are the reduced Hessian blocks, i.e. the pivot blocks of $\tilde{M}_i$. And the equivalence of their positive definiteness is thus established by Sylvester's law of inertia, i.e. $\tilde{M}_i \succ 0$ if and only if every $\tilde{M}_k \succ 0$, which indicates the Cholesky factorizations of all $\tilde{M}_k$ in the backward pass succeed. We can, therefore, have Riccati recursion as the indefiniteness detection, as also used in Fatrop [22].

3.2.2 Stage regularization in Riccati recursion

With this connection, we can regularize the reduced Hessian $\tilde{M}_i$, while preserving the sparse structure of the KKT system, by a stage-wise regularization strategy applied to each reduced Hessian block $\tilde{M}_k$ during the Riccati recursion.

Using the failure of Cholesky factorization as the sign, we can regularize the reduced Hessian block $\tilde{M}_k$ on the fly by adding a block adjustment matrix E_k to it, i.e. $\tilde{M}_k + E_k$. This gives a modification of the backward pass of Riccati recursion as follows:

Algorithm 4 Backward Pass of Modified Riccati Recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\text{stat},x}$

for $k = N-1$ **down to** 0 **do**
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P_{k+1} B_k$
  $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  if factorization fails **then**
    $\tilde{M}_k \leftarrow \tilde{M}_k + E_k$ ▷ Apply regularization
    $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  end if
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P_{k+1} A_k$

  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}^k_{\text{stat},u} + B_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) \right)$

  $P_k \leftarrow \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}^k_{\text{stat},x} + A_k^\top (P_{k+1} r^k_{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

The regularization is thus referred to as Riccati-based regularization and is only applied to the stages where the reduced Hessian block is indefinite.

3.3 Improved stage regularization strategy

In this section, we will analyze the case where the commonly used minimum perturbation strategy in regularization fails in stage regularization, which would

cause a severe condition number issue in the preceding stages due to the propagation of the negative curvature through the stages. This motivates us to propose an improved stage regularization strategy that avoids minimum perturbation and takes the propagation into account.

3.3.1 Failure of minimum perturbation regularization

In this subsection, we show with a general example that the minimum perturbation strategy, e.g. PROJECT, which is commonly the first choice because it modifies the original problem the least, can cause a condition number issue in the stage regularization.

Suppose at stage k, the reduced Hessian block $\tilde{M}_k$ is not PD. Without loss of generality, we can assume $\tilde{M}_k$ has only one negative eigenvalue ν_j, and the corresponding eigenvector is v_j. The minimum perturbation strategy is applied as:

$$E_k = (-\nu_j + \epsilon) v_j v_j^\top, \tag{3.6}$$

where ϵ is a small positive value to ensure the positive definiteness of the regularized reduced Hessian block.

By the Sherman–Morrison formula, the inverse of the regularized reduced Hessian block can be expressed as:

$$\begin{aligned}(\tilde{M}_k + E_k)^{-1} &= \tilde{M}_k^{-1} + \frac{\nu_j - \epsilon}{\epsilon \nu_j} v_j v_j^\top \\ &= \tilde{M}_k^{-1} + (\frac{1}{\epsilon} - \frac{1}{\nu_j}) v_j v_j^\top,\end{aligned}$$

and the update of the cost-to-go matrix P_k follows as:

$$\begin{aligned}P_k &= \tilde{Q}_k + A_k^\top P_{k+1} A_k + \Gamma_k^\top K_k \\ &= \tilde{Q}_k + A_k^\top P_{k+1} A_k - \Gamma_k^\top \tilde{M}_k^{-1} \Gamma_k - (\frac{1}{\epsilon} - \frac{1}{\nu_j}) \Gamma_k^\top v_j v_j^\top \Gamma_k \\ &= P_k^{\text{unreg}} - (\frac{1}{\epsilon} - \frac{1}{\nu_j}) \Gamma_k^\top v_j v_j^\top \Gamma_k.\end{aligned} \tag{3.7}$$

Note that this negative rank-one modification can already render P_k itself indefi-

nite, independently of its downstream effect on $\tilde{M}_{k-1}$. And this can happen even when the perturbation on $\tilde{M}_k$ is moderate, provided that the negative-curvature direction v_j is strongly coupled to the state, i.e. $\|\Gamma_k^\top v_j\|$ is large.

Consequently, the reduced Hessian block $\tilde{M}_{k-1}$ at the previous stage $k-1$ is calculated as:

$$\tilde{M}_{k-1} = \tilde{M}_{k-1}^{\text{unreg}} - (\frac{1}{\epsilon} - \frac{1}{\nu_j})(B_{k-1}^\top \Gamma_k^\top v_j)(B_{k-1}^\top \Gamma_k^\top v_j)^\top, \tag{3.8}$$

and the condition number of the regularized reduced Hessian block $\tilde{M}_{k-1}$ can be approximated as:

$$\kappa(\tilde{M}_{k-1}) \approx \frac{c(\epsilon + |\nu_j|)}{\epsilon|\nu_j|}, \tag{3.9}$$

where $c = \frac{\|B_{k-1}^\top \Gamma_k^\top v_j\|^2}{\nu_{\min}(\tilde{M}_{k-1}^{\text{unreg}})}$ is an $O(1)$ constant.

As ϵ is a small positive value, the presence of the negative rank-one update, scaled by roughly $\frac{1}{\epsilon}$, will dominate, not only severely deteriorating its condition number, but also likely making $\tilde{M}_{k-1}$ indefinite, which can further cause a domino effect through the stages.

3.3.2 Insights and improved stage regularization

This issue reveals two insights into stage regularization:

- The minimum modification objective [23] is not the foremost consideration for stage regularization, as it can cause a severe condition number issue in the preceding stages. A trade-off between perturbation norm and condition number must be carefully treated.
- The cost-to-go matrix P_k must be incorporated into the regularization design, as the severity of propagation depends on the coupling structure between stages, which a purely local scheme cannot capture.

From the first insight, instead of the minimum perturbation strategy, an intuitive, but effective, approach is to use the absolute value of the negative eigenvalues or $\text{LDL}^\top$ pivots as the perturbation magnitude, which can avoid the severe condition number issue in the preceding stages. When it comes to the EM strategy, this is

the MIRROR strategy, and when it comes to the MC strategy, this is the Type-II strategy [23], of which GMW is a common representative.

From the second insight, we can incorporate the cost-to-go matrix P_k into the strategy design, which also can be achieved by adding an adjustment matrix $E^i_{k+1,P}$ to the cost-to-go matrix P_{k+1} directly, so that the propagation of the negative curvature can be taken into account in the regularization design.

The resulting improved stage regularization strategy can be summarized as follows:

Algorithm 5 Backward Pass of Improved Modified Riccati Recursion

Initialize: $P_N \leftarrow \tilde{Q}_N$, $p_N \leftarrow \tilde{r}^N_{\text{stat},x}$, $\mathcal{E}(\cdot)$: MIRROR or Type-II strategy

for $k = N-1$ **down to** 0 **do**
  $P^{\text{reg}}_{k+1} \leftarrow P_{k+1}$ ▷ Default with no regularization
  $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P^{\text{reg}}_{k+1} B_k$
  $L_k \leftarrow \text{Cholesky}(\tilde{M}_k)$
  if factorization fails **then**
    $P^{\text{reg}}_{k+1} \leftarrow \mathcal{E}(P_{k+1})$ ▷ Regularize cost-to-go matrix
    $\tilde{M}_k \leftarrow \tilde{R}_k + B_k^\top P^{\text{reg}}_{k+1} B_k$
    $\tilde{M}^{\text{reg}}_k \leftarrow \mathcal{E}(\tilde{M}_k)$ ▷ Regularize reduced Hessian block
    $L_k \leftarrow \text{Cholesky}(\tilde{M}^{\text{reg}}_k)$
  end if
  $\Gamma_k \leftarrow \tilde{S}_k + B_k^\top P^{\text{reg}}_{k+1} A_k$
  $K_k \leftarrow -L_k^{-\top} L_k^{-1} \Gamma_k$
  $k_k \leftarrow -L_k^{-\top} L_k^{-1} \left(\tilde{r}^k_{\text{stat},u} + B_k^\top (P^{\text{reg}}_{k+1} r^k_{\text{eq}} + p_{k+1}) \right)$
  $P_k \leftarrow \tilde{Q}_k + A_k^\top P^{\text{reg}}_{k+1} A_k + \Gamma_k^\top K_k$
  $p_k \leftarrow \tilde{r}^k_{\text{stat},x} + A_k^\top (P^{\text{reg}}_{k+1} r^k_{\text{eq}} + p_{k+1}) + \Gamma_k^\top k_k$
end for

In Algorithm 5, when the Cholesky factorization fails, the strategy first applies $\mathcal{E}(\cdot)$ to the cost-to-go matrix, which stops the negative-curvature propagation identified in the second insight, and then applies it again to the reduced Hessian block to act as a local safeguard, ensuring that the Cholesky factorization succeeds.

Both corrections are produced by the same non-minimal strategy $\mathcal{E}(\cdot)$, MIRROR or Type-II, rather than by the minimum modification rule. This keeps the condi-

tion number of the regularized blocks bounded, as motivated by the first insight. When no factorization failure occurs, the recursion reduces to the standard Riccati backward pass.

3.4 Asymptotic inactivity of the regularization

Since the regularization is applied only as the detection in Section 3.2.1 triggers it, its behavior close to the solution is governed by the reduced Hessian $\tilde{M}_i$. In this section, we prove that any regularization triggered by this detection is asymptotically inactive, i.e., it vanishes once the iterates are sufficiently close to the solution. The argument adapts that of [24] to the reduced Hessian $\tilde{M}_i$.

We will start from the system in Equation (2.9) and make the following assumptions for the proof:

Assumption 1. *(y^*, λ^*) is a local (or global) minimizer for the QP in Equation* (2.2)*.*

Assumption 2. *Each local minimizer (y^*, λ^*) for the QP in Equation* (2.2) *is isolated.*

Assumption 3. *LICQ holds at the minimizer (y^*, λ^*).*

Assumption 4. *Strict complementarity holds at the minimizer (y^*, λ^*).*

Assumptions 1–3 imply the second-order sufficient condition (SOSC) for local optimality, so we have the following Lemma 3:

Lemma 3. *Under Assumptions 1–3, for all $w \in C(y^*, \lambda^*) \setminus \{0\}$, it holds that $w^\top H w > 0$.*

Proof. With Assumption 1, the second-order necessary condition (SONC) for local optimality holds, which states that for all $w \in C(y^*, \lambda^*)$, we have $w^\top H w \geq 0$.

We proceed by contradiction. Suppose there exists $w \in C(y^*, \lambda^*) \setminus \{0\}$ such that $w^\top H w = 0$. Consider the point $y(\alpha) = y^* + \alpha w$ for $\alpha > 0$. Because the constraints are linear and w is in the critical cone, $y(\alpha)$ remains feasible for sufficiently small α.

Since the objective function of Equation (2.2) is quadratic, expanding $f(y(\alpha))$ yields:

$$f(y(\alpha)) = f(y^*) + \alpha \nabla f(y^*)^\top w + \frac{1}{2}\alpha^2 w^\top H w$$

As $\nabla f(y^*)^\top w = 0$, given the assumption that $w^\top H w = 0$, the equation reduces exactly to $f(y(\alpha)) = f(y^*)$. This implies that $y(\alpha)$ is also a local minimizer, which directly contradicts Assumption 2. Therefore, $w^\top H w > 0$ must hold. □

Proposition 4. *Under Assumptions 1–4, let the regularization be triggered by the Riccati-based detection. As* $\{(y_i, \lambda_i)\} \to (y^*, \lambda^*)$, $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$ *and hence* $E_i = 0$ *for all* i *sufficiently large.*

Proof. Under Assumption 4, the critical cone $C(y^*, \lambda^*)$ reduces to the tangent space of the active constraints. Therefore, we have $C(y^*, \lambda^*) = \{w \mid Fw = 0,\ G_{\mathcal{A}} w = 0\}$.

For any v with $\|v\| = 1$, write $w = Zv \in \{w \mid Fw = 0\} \setminus \{0\}$ and let G_j denote the j-th row of G. The corresponding quadratic form of the reduced Hessian reads:

$$\begin{aligned} v^\top \tilde{M}_i v &= w^\top (H + G^\top T_i^{-1} \Lambda_i G) w \\ &= w^\top H w + \sum_j \frac{\lambda_j^i}{t_j^i} (G_j w)^2 \\ &= w^\top H w + \sum_{j \in \mathcal{A}} \frac{\lambda_j^i}{t_j^i} (G_j w)^2 + \sum_{j \notin \mathcal{A}} \frac{\lambda_j^i}{t_j^i} (G_j w)^2 \end{aligned}$$

We now proceed by contradiction. Assume that $\tilde{M}_i$ is not PD for infinitely many i. Then, passing to a subsequence, there exist v_i with $\|v_i\| = 1$ and $v_i \to \bar{v}$ such that $v_i^\top \tilde{M}_i v_i \leq 0$ for all i in this subsequence. Writing $w_i = Z v_i$ and $\bar{w} = Z\bar{v}$, so that $w_i \to \bar{w}$, we can distinguish two cases:

For $G_{\mathcal{A}}\bar{w} = 0$, the limit direction satisfies $\bar{w} \in C(y^*, \lambda^*) \setminus \{0\}$, so $\bar{w}^\top H \bar{w} > 0$ follows by Lemma 3. Since the remaining terms are nonnegative, $v_i^\top \tilde{M}_i v_i \geq w_i^\top H w_i$ with $w_i^\top H w_i \to \bar{w}^\top H \bar{w} > 0$. Hence $v_i^\top \tilde{M}_i v_i > 0$ for all sufficiently large i, contradicting the assumption.

For $G_{\mathcal{A}}\bar{w} \neq 0$, Assumption 4 gives $\lambda_j^i \to \lambda_j^* > 0$ and $t_j^i \to 0$ for $j \in \mathcal{A}$, so the term $\sum_{j\in\mathcal{A}} \frac{\lambda_j^i}{t_j^i}(G_j w_i)^2 \to \infty$, which dominates the other two terms, and thus $v_i^\top \tilde{M}_i v_i \to \infty$, contradicting the assumption.

Therefore, we can conclude that $\tilde{M}_i = Z^\top \tilde{H}_i Z \succ 0$ for all i sufficiently large. By Section 3.2.1, all Cholesky factorizations in the backward pass then succeed and $E_i = 0$. □

Proposition 4 shows the regularization triggered by the Riccati-based detection is asymptotically inactive: once the iterates are close to (y^*, λ^*), $E_i = 0$ and the regularized recursion coincides with the unregularized one. The solver thus recovers the exact Newton step and inherits the local convergence rate of the underlying IPM, without the regularization affecting its asymptotic behavior.

Chapter 4

A benchmark for indefinite OCP QPs

In order to evaluate the performance of the proposed Riccati-based primal regularization, especially in the context of the subproblems of SQP, in this chapter, we aim at constructing a benchmark of indefinite OCP QPs that arise in the SQP iterations of NMPC problems. We will first introduce the construction of the QP benchmark, including the formulation of QP subproblems in the benchmark, an overview of OCP test problems and the pipeline for collecting indefinite QPs in Section 4.1. Then we will analyze the collected problem instances to gain a better understanding of characteristics regarding definiteness in the indefinite OCP QP benchmark in Section 4.2.

4.1 Benchmark construction

This section describes the construction of the indefinite QP benchmark. We first formulate the QP subproblems from the SQP iterations of the OCPs, then introduce the eight OCP test problems, formulated as fixed-time regulator problems with quadratic stage costs and box constraints on the controls, and finally the collection pipeline.

4.1.1 Formulation of QP subproblems from OCPs

We will start from the formulation of the nonlinear OCPs to obtain the resulting QP subproblems that arise in the SQP iterations. In the benchmark, all OCPs share the same structure: a fixed-horizon, direct multiple-shooting OCP with a quadratic stage cost that penalizes deviations from given state and control references. The mathematical formulation of the OCPs is given as follows:

$$\begin{aligned}
\underset{x,\,u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \frac{1}{2} \begin{bmatrix} x_k - x_k^{\text{ref}} \\ u_k - u_k^{\text{ref}} \end{bmatrix}^\top \begin{bmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{bmatrix} \begin{bmatrix} x_k - x_k^{\text{ref}} \\ u_k - u_k^{\text{ref}} \end{bmatrix} \\
& + \tfrac{1}{2} \| x_N - x_N^{\text{ref}} \|_{\mathcal{Q}_N}^2 \\
\text{subject to} \quad & x_0 = \bar{x}_0, \\
& x_{k+1} = f_k(x_k, u_k), \qquad k = 0, \ldots, N-1, \\
& \underline{g} \leq \mathcal{C}_k x_k + \mathcal{D}_k u_k \leq \bar{g}, k = 0, \ldots, N-1, \\
& \underline{h} \leq h_k(x_k, u_k) \leq \bar{h}, \quad k = 0, \ldots, N-1, \\
& \underline{g}_N \leq \mathcal{C}_N x_N \leq \bar{g}_N,
\end{aligned} \tag{4.1}$$

where $x_{k+1} = f_k(x_k, u_k)$ is one integration step of the continuous dynamics over $[t_k, t_{k+1}]$. $\bar{x}_0$ is the given initial state, $\mathcal{Q}_k, \mathcal{R}_k, \mathcal{Q}_N \succeq 0$ are weighting matrices, $\mathcal{S}_k$ is the coupling cost matrix, and $x_k^{\text{ref}}, u_k^{\text{ref}}$ are the reference points. $\mathcal{C}_k x_k + \mathcal{D}_k u_k$ includes the affine and box path constraints, $h_k(\cdot, \cdot)$ collects the nonlinear path constraints, and $\underline{g}_N \leq \mathcal{C}_N x_N \leq \bar{g}_N$ are the terminal inequality constraints.

When using the exact Hessian of the Lagrangian in the SQP iterations, the resulting Hessian of the QP subproblem at stage $k = 0, \ldots, N-1$ reads:

$$\begin{aligned}
H_k &= \nabla^2 \mathcal{L}(x, u, \pi, \lambda) \\
&= \begin{bmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{bmatrix} + \sum_{j=1}^{n_x} (\pi_k)_j \nabla^2 [f_k]_j(x_k, u_k) + \sum_{j=1}^{n_h} (\lambda_k)_j \nabla^2 [h_k]_j(x_k, u_k),
\end{aligned} \tag{4.2}$$

where n_h is the number of nonlinear constraints, and π_k and λ_k are the Lagrange multipliers associated with the dynamics and nonlinear constraints, respectively.

The definiteness of H_k is thus determined by two parts, a constant cost block $\left[\begin{smallmatrix} \mathcal{Q}_k & \mathcal{S}_k^\top \\ \mathcal{S}_k & \mathcal{R}_k \end{smallmatrix}\right]$ and an iterate-dependent cost from the two curvature terms. In the

latter, each component Hessian $\nabla^2[f_k]_j$ and $\nabla^2[h_k]_j$ is symmetric but carries no definiteness guarantee, and the associated multipliers $(\pi_k)_j$ and $(\lambda_k)_j$ may take arbitrary sign and magnitude during the SQP iterations.

The weighted sum H_k is thus a general symmetric and typically indefinite matrix, in particular when the multipliers are large. Consequently, the exact-Hessian QP subproblem is in general nonconvex.

As for H_N, neither the dynamics nor the affine terminal constraints add curvature in x_N, so $H_N = \mathcal{Q}_N \succeq 0$ is constant.

Collecting the stage Hessians H_k and Jacobians and linearizing the constraints of Equation (4.1) at the current SQP iterate yields a structured QP. The subproblem takes the form:

$$
\begin{aligned}
\underset{\Delta x,\,\Delta u}{\text{minimize}} \quad & \sum_{k=0}^{N-1} \left(\frac{1}{2} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix}^\top \begin{bmatrix} Q_k & S_k^\top \\ S_k & R_k \end{bmatrix} \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix} + \begin{bmatrix} q_k \\ r_k \end{bmatrix}^\top \begin{bmatrix} \Delta x_k \\ \Delta u_k \end{bmatrix} \right) \\
& \quad + \tfrac{1}{2}\, \Delta x_N^\top Q_N\, \Delta x_N + q_N^\top \Delta x_N \\
\text{subject to} \quad & \Delta x_0 = \Delta \bar{x}_0, \\
& \Delta x_{k+1} = A_k \Delta x_k + B_k \Delta u_k + c_k, \qquad k = 0, \dots, N-1, \\
& \begin{bmatrix} \underline{u}_k \\ \underline{x}_k \\ \underline{d}_k \end{bmatrix} \leq \begin{bmatrix} J_k^{b,u} & 0 \\ 0 & J_k^{b,x} \\ D_k & C_k \end{bmatrix} \begin{bmatrix} \Delta u_k \\ \Delta x_k \end{bmatrix} \leq \begin{bmatrix} \overline{u}_k \\ \overline{x}_k \\ \overline{d}_k \end{bmatrix}, k = 0, \dots, N-1, \\
& \underline{\Delta x}_N \leq J_N^{b,x} \Delta x_N \leq \overline{\Delta x}_N,
\end{aligned}
\tag{4.3}
$$

where, at each stage k, $\Delta x_k \in \mathbb{R}^{n_x}$ and $\Delta u_k \in \mathbb{R}^{n_u}$ denote the state and control deviation variables at stage k. Note that Δx_k and Δu_k here denote the SQP step, which the QP solver treats as its decision variables x_k and u_k in Equation (2.10).

All remaining quantities are obtained by linearizing the OCP at the current SQP iterate (x_k, u_k) with multipliers (π_k, λ_k).

In the objective, Q_k, R_k and S_k are the state, control and coupling blocks of the exact Lagrangian Hessian H_k, and the stage-cost gradients are $q_k = \mathcal{Q}_k(x_k - x_k^{\text{ref}}) + \mathcal{S}_k^\top(u_k - u_k^{\text{ref}})$ and $r_k = \mathcal{R}_k(u_k - u_k^{\text{ref}}) + \mathcal{S}_k(x_k - x_k^{\text{ref}})$, with Q_N and q_N as the terminal Hessian and gradient.

In the equality constraints, $A_k = \nabla_{x_k} f_k$ and $B_k = \nabla_{u_k} f_k$ are the Jacobians of the discretized dynamics, $c_k = f_k(x_k, u_k) - x_{k+1}$ is the dynamics residual, which vanishes at a dynamically feasible iterate, and $\Delta\bar{x}_0 = \bar{x}_0 - x_0$ is the residual of the initial state constraint $x_0 = \bar{x}_0$.

In the inequality constraints, $J_k^{b,u}$ and $J_k^{b,x}$ are selection matrices that pick out the bounded controls and states, with bounds $\underline{u}_k, \overline{u}_k$ and $\underline{x}_k, \overline{x}_k$. The matrices $C_k = \begin{bmatrix} \mathcal{C}_k \\ \nabla_{x_k} h_k \end{bmatrix}$ and $D_k = \begin{bmatrix} \mathcal{D}_k \\ \nabla_{u_k} h_k \end{bmatrix}$ stack the Jacobians of the affine and nonlinear path constraints, with the bounds

$$\underline{d}_k = \begin{bmatrix} \underline{g} - \mathcal{C}_k x_k - \mathcal{D}_k u_k \\ \underline{h} - h_k(x_k, u_k) \end{bmatrix}, \qquad \overline{d}_k = \begin{bmatrix} \overline{g} - \mathcal{C}_k x_k - \mathcal{D}_k u_k \\ \overline{h} - h_k(x_k, u_k) \end{bmatrix}.$$

4.1.2 Benchmark OCPs

In this subsection, we introduce the eight OCPs used to construct the benchmark. Each OCP is based on a dynamic system provided in [25] and formulated as a regulation or point-to-point transfer problem with problem-specific modifications, which can be uniformly described with Equation (4.1).

The eight OCPs are named and characterized as follows:

- **Pendulum:** Swing-up of an inverted pendulum on a cart under an input bound and a mixed state–input constraint.
- **Acrobot:** Adapted from [26]. An underactuated two-link pendulum moved between two prescribed configurations.
- **Dubins car:** Adapted from [27]. Parallel parking of a kinematic unicycle in a narrow corridor.
- **Furuta:** Adapted from [28]. Swing-up of a rotary inverted pendulum under tight torque limits.
- **Flying robot:** Adapted from [29]. A planar free-flying robot steered by four nonnegative thrusters.
- **Robot arm:** Adapted from [29]. A three-degree-of-freedom arm moved to a new set-point, with nonlinear input coupling.
- **Crane:** Adapted from [29]. An overhead crane moving a load between two rest positions.

- **Moonlander:** Adapted from [30]. Soft landing of a planar lunar lander with two thrusters.

Table 4.1: Summary of the evaluated OCPs in common control problem scales.

Problem	n_x	n_u	N
Pendulum	4	1	25
Acrobot	4	1	15
Dubins car	3	2	25
Furuta	4	1	20
Flying robot	6	4	50
Robot arm	6	3	30
Crane	6	2	40
Moonlander	6	2	30

Besides the different physical systems, these OCPs also cover different state and control dimensions, as shown in Table 4.1. The state dimension n_x ranges from 3 to 6, the control dimension n_u from 1 to 4, and the horizon length N from 15 to 50. This spread is chosen so that the benchmark spans a range of common problem scales in control problems.

4.1.3 Collection pipeline

Based on the eight OCPs, we can construct a benchmark of indefinite QPs arising from SQP-based NMPC. Specifically, we establish a collection pipeline that extracts the QP subproblems directly from intermediate SQP iterations. The collection pipeline consists of two steps: *generate* and *collect*, and can be summarized as follows:

1. **Generate:** In this step, we generate iterates by solving the OCP with SQP and obtain iterates from early to late SQP iterations. To keep the SQP iterations from failing on indefinite exact-Hessian QPs, we use SQP-level regularization or the Gauss–Newton Hessian approximation in `acados`.
2. **Collect:** In this step, we utilize the previously generated iterates as initial guesses for linearization within the solver. We can selectively collect indefinite QPs that exhibit an indefinite reduced Hessian.

In the collect step, HPIPM uses the classical Riccati recursion (`ric_alg = 0`),

which requires only the reduced Hessian to be PD, instead of the square-root version [31], which requires the full Hessian to be PD. A failure of the recursion thus indicates an indefinite reduced Hessian, which `acados` reports as status `4` (infeasible QP), and only the QPs with this status are collected.

For each problem, we limit the number of near-identical QPs, so that the number of instances per problem stays within a comparable range and the collected instances remain meaningful. We also use IPOPT [16] as a reference solver and keep only the QPs that it solves successfully. After this filtering, we obtain a total of 175 indefinite QPs from the eight OCPs.

The per-problem counts are reported in Table 4.2 and range from 15 to 29. These QPs are stored in JSON format and are available at `https://github.com/Pandatheon/Primal-dual_IPM.git`.

Table 4.2: Number of indefinite QPs collected from each OCP (175 in total).

Problem	**Number**
Pendulum	18
Acrobot	15
Dubins car	29
Furuta	19
Flying robot	20
Robot arm	24
Crane	29
Moonlander	21

4.2 Benchmark analysis

In this section, we will analyze the collected instances regarding the definiteness in the benchmark. The distribution of indefiniteness is analyzed in Section 4.2.1, which describes the magnitude and coverage of the indefinite reduced Hessians in the collected QPs, and the source of indefiniteness is analyzed in Section 4.2.2, which further illustrates the source of indefiniteness in the collected QPs by analyzing the eigenvalues of the reduced Hessian blocks at each stage.

4.2.1 Distribution of indefiniteness

We first characterize the collected QPs along two axes: the *depth* of indefiniteness, measured by the magnitude of the most negative eigenvalue $|\nu_{\min}|$ of the reduced Hessian, and the *width*, measured by the fraction of reduced Hessian blocks that are indefinite within a QP.

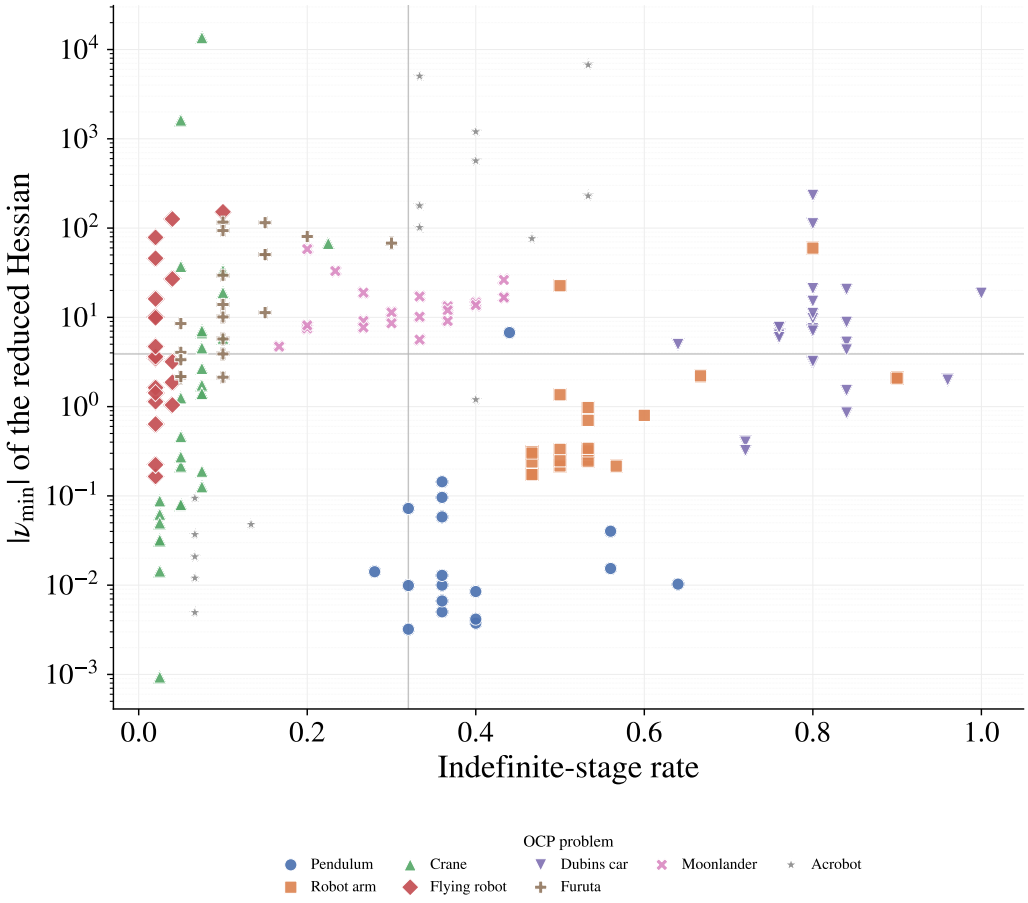


Figure 4.1: Depth ($|\nu_{\min}|$ of the reduced Hessian, log scale) versus width (indefinite-stage rate) of the collected QPs, colored by OCP.

As seen from Figure 4.1, the depth of the collected QPs ranges from near zero to very large values, and each OCP occupies its own region of the depth–width plane. Problems that are indefinite in only a small fraction of their stages, such as *Crane*, *Flying robot* and *Furuta* (indefinite-stage rates below 0.30), can still reach deep negative curvature. Conversely, *Dubins car* and *Robot arm*, with indefinite-stage rates from 0.47 up to 1, show only moderate depth. In the collected QPs, wide and deep indefiniteness thus seldom occur together.

Moonlander and *Acrobot* lie in between, with intermediate indefinite-stage rates. *Moonlander* stays within a narrow depth band, whereas *Acrobot* spans almost the full depth range of the benchmark. *Pendulum* contributes QPs with very shallow negative curvature at intermediate rates. The eight OCPs thus cover narrow-but-deep, intermediate and wide-but-moderate indefiniteness.

4.2.2 Source of indefiniteness

Recall that the reduced Hessian block at stage k, $M_k = R_k + B_k^\top P_{k+1} B_k$, is assembled from two parts, the control cost block R and the cost-to-go block P.

To investigate where the indefiniteness originates, we classify every indefinite stage by which of R_k and P_{k+1} has a negative eigenvalue: *R only* ($R_k \not\succeq 0$, $P_{k+1} \succeq 0$), *P only* ($P_{k+1} \not\succeq 0$, $R_k \succeq 0$), and *both* ($R_k, P_{k+1} \not\succeq 0$). Since $R_k, P_{k+1} \succeq 0$ would imply $M_k \succeq 0$, these three cases cover all indefinite stages.

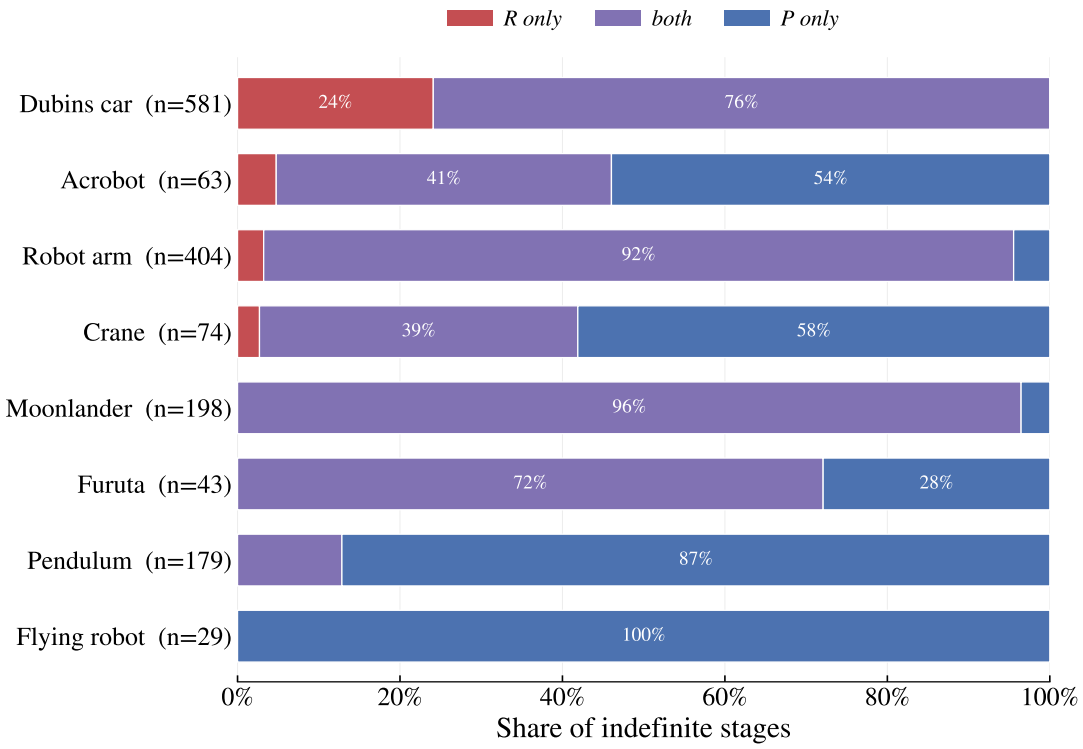


Figure 4.2: Per-problem composition of stage indefiniteness. Each bar shows, for one OCP, the share of stages that are *R only*, *both* and *P only* (n is the number of collected indefinite stages).

Figure 4.2 shows this classification for each OCP. In every OCP, most indefinite stages involve P, either alone or together with R, and six of the eight OCPs are dominated by a single category ($\geq 70\%$ of their stages), with only *Acrobot* and *Crane* mixed.

The *both* case dominates *Moonlander* (96%), *Robot arm* (92%), *Dubins car* (76%) and *Furuta* (72%). The *P only* case dominates *Flying robot* (100%) and *Pendulum* (87%), and is also the largest category in the two mixed problems, *Crane* (58% against 39% *both*) and *Acrobot* (54% against 41% *both*). The *R only* case appears in four problems and is substantial only for *Dubins car* (24% of its stages), and in *Robot arm*, *Acrobot* and *Crane* it stays below 5%.

Among all, *Flying robot* is the clearest case: all of its indefinite stages are *P only*, with $R_k \succ 0$ throughout. Its negative curvature thus enters the reduced Hessian only through the cost-to-go block, at isolated stages.

Chapter 5

Numerical experiments

In this chapter, we evaluate the proposed regularization methods numerically and discuss the results. We begin in Section 5.1 with the two metrics used throughout the comparison, the iteration profile and the flop unit estimation, which measure convergence behavior and factorization cost respectively. Section 5.2 then describes the prototype solver on which all experiments are run and stresses that it reproduces the iteration behavior of HPIPM on convex QPs, so that any difference observed later can be attributed to the regularization rather than to the solver itself. Section 5.3 fixes the experiment setup, naming and classifying the regularization methods under comparison and stating the parameters shared by all of them. Section 5.4 presents the numerical comparison, first among the Riccati-based methods themselves and then against full Hessian regularization and inertia correction, followed by ablations of the barrier update and the line search. Section 5.5 concludes the chapter by analyzing the failures and the Mehrotra ablation.

5.1 Experiment metrics

In this section, we will introduce the metrics used to evaluate the performance of different regularization strategies in our experiments. Since we care about both the convergence behavior and the factorization cost caused by different regularization methods, we will use two complementary metrics: iteration profile and flop unit estimation. The former one, iteration profile, is used to evaluate how

many iterations each method takes to converge on a set of test instances, while the latter one, flop unit estimation, is approximated by the number of modified Cholesky factorizations of the reduced Hessian, which is used to evaluate the factorization cost of each method.

5.1.1 Iteration profile

Let $\mathcal{P}$ denote the set of test instances and $\mathcal{M}$ the set of candidate regularization strategies. For a method $m \in \mathcal{M}$ and an instance $\mathrm{p} \in \mathcal{P}$, let $i_{m,\mathrm{p}}$ be the number of iterations m needs to satisfy the convergence criterion on p. If the run reaches the iteration cap $i_{\max}$, the instance is counted as unsolved and $i_{m,\mathrm{p}} = \infty$. The *iteration profile* of method m is

$$\Pi_m(\ell) \;=\; \Big|\{\, \mathrm{p} \in \mathcal{P} : i_{m,\mathrm{p}} \le \ell \,\}\Big|, \qquad \ell = 1, \dots, i_{\max}, \tag{5.1}$$

i.e. the number of instances that m solves within ℓ iterations.

A curve that rises steeply at small ℓ indicates an *efficient* method, and its final value $\Pi_m(i_{\max})$, the number of solved instances, measures its *robustness*.

5.1.2 Flop unit estimation

Iteration counts alone do not reflect the factorization cost performed per iteration, which differs substantially across regularization strategies, where some are based on eigenvalue decomposition, some are based on modified Cholesky factorization, and some only repeat the Cholesky factorization with an increasing diagonal shift. To compare methods on a common scale, we take a proxy unit for unifying the factorization cost.

We take one Cholesky factorization of the block reduced Hessian as the proxy unit. For a control dimension n_u, factorizing an $n_u \times n_u$ matrix costs approximately $\frac{1}{3}n_u^3$ flops, hence

$$\text{unit} \;\approx\; \tfrac{1}{3}\, n_u^3. \tag{5.2}$$

The modified factorization only adds diagonal updates of $\mathcal{O}(n_u^2)$ cost, so to leading

order it also costs one unit.

The other factorizations and decompositions are expressed relative to this unit. A symmetric eigenvalue decomposition of the same block costs about $9n_u^3$ flops [32], i.e.

$$\frac{9n_u^3}{\frac{1}{3}n_u^3} \;=\; 27 \text{ units.} \tag{5.3}$$

Regarding the cost-to-go matrix $P_{k+1} \in \mathbb{R}^{n_x \times n_x}$ or a full Hessian block $\tilde{H}_k \in \mathbb{R}^{(n_x+n_u)\times(n_x+n_u)}$, the ratios are considered as:

$$\omega_P \;=\; \left(\frac{n_x}{n_u}\right)^3, \quad \omega_H \;=\; \left(\frac{n_x+n_u}{n_u}\right)^3. \tag{5.4}$$

Thus, a modified Cholesky factorization of P_{k+1} costs ω_P units and an eigenvalue decomposition $27\omega_P$ units, and likewise ω_H and $27\omega_H$ units for $\tilde{H}_k$. And since the full Hessian regularization modifies all N blocks $\tilde{H}_k$ at once, each regularization costs $N\omega_H$ or $27N\omega_H$ units.

5.2 Prototype solver implementation

To ensure a fair comparison, we implemented all regularization methods in one prototype solver. The Riccati-based primal–dual IPM prototype solver is implemented in Python, using HPIPM as a reference. The prototype was designed to reproduce its iteration behavior on convex QPs and is available at `https://github.com/Pandatheon/Primal-dual_IPM.git`.

One implementation detail deserves mention, as it deviates from the plain FTB rule stated in Section 2.4. Rather than scaling the maximal step by a fixed fraction η, the primal and dual step sizes α_{prim} and α_{dual} are obtained from their respective maximal step through the heuristic

$$\alpha = \begin{cases} \alpha_{\max}\Big[(1-\alpha_{\max})\,\kappa_\eta + \alpha_{\max}\hat{\eta}\Big], & \alpha_{\max} < 1, \\ \alpha_{\max}, & \alpha_{\max} = 1, \end{cases} \tag{5.5}$$

where $\alpha_{\max} \in (0, 1]$ denotes the maximal step length from Equation (2.22) or

Equation (2.23), $\kappa_\eta = 0.99$ and $\hat{\eta} = 1 - 10^{-7}$.

When $\alpha_{\max} < 1$, the bracketed factor changes with $\alpha_{\max}$, from a strongly blocked step to a nearly full one, so the closer the maximal step is to a full step, the less it is shortened. When $\alpha_{\max} = 1$, an unblocked full step can be taken without scaling by $\hat{\eta}$ so that the local quadratic convergence rate of the IPM can be preserved.

We further align the prototype solver with HPIPM in terms of the mode settings by defining a set of HPIPM settings, referred to in the following as the `CUSTOMIZED` mode, whose values are listed in Table 5.1.

Table 5.1: Detailed settings for the defined `CUSTOMIZED` mode.

Parameter	**Value**	**Parameter**	**Value**
mu0	1e1	lam_min	1e-12
alpha_min	0.0	t_min	1e-12
res_g_max	1e-6	tau_min	1e-12
res_b_max	1e-8	lam0_min	1e-12
res_d_max	1e-8	t0_min	1e-12
res_m_max	1e-6	warm_start	0
dual_gap_max	1e15	abs_form	0
iter_max	30	comp_dual_sol_eq	1
stat_max	30	comp_res_exit	1
pred_corr	1	comp_res_pred	1
cond_pred_corr	1	split_step	1
itref_pred_max	0	m_safe	0.5
itref_corr_max	0	var_init_scheme	0
reg_prim	0.0	t_lam_min	2
square_root_alg	0	t0_init	2
lq_fact	0	update_fact_exit	1

As one can notice, the `CUSTOMIZED` mode largely coincides with the `SPEED` mode of the original HPIPM, differing in three respects. First, the square-root algorithm is disabled (`square_root_alg`= 0), so that the Riccati recursion requires no PD full Hessian to carry on, second, the primal and dual step sizes are computed separately (`split_step`= 1) instead of sharing a single step length, third, the minimal step size is disabled by setting `alpha_min`= 0. In both modes no iterative refinement is applied (`itref_pred_max`=`itref_corr_max`= 0) and the conditional Mehrotra's predictor–corrector is retained (`pred_corr`=`cond_pred_corr`= 1).

To verify that the two implementations do agree, we collected 40 convex QP

subproblems, five from each of the 8 OCPs in the benchmark, and compared the iteration behavior of the prototype solver against that of `CUSTOMIZED`-mode HPIPM. The resulting iteration profiles are shown in Figure 5.1.

Both solvers solve all 40 instances in the exact same number of iterations for each instance, resulting in iteration profiles that coincide. On convex QPs, the prototype solver therefore reproduces the iteration behavior of HPIPM, which verifies the prototype as a reliable baseline and allows us to attribute the differences observed in the experiments to the regularization rather than to the solver.

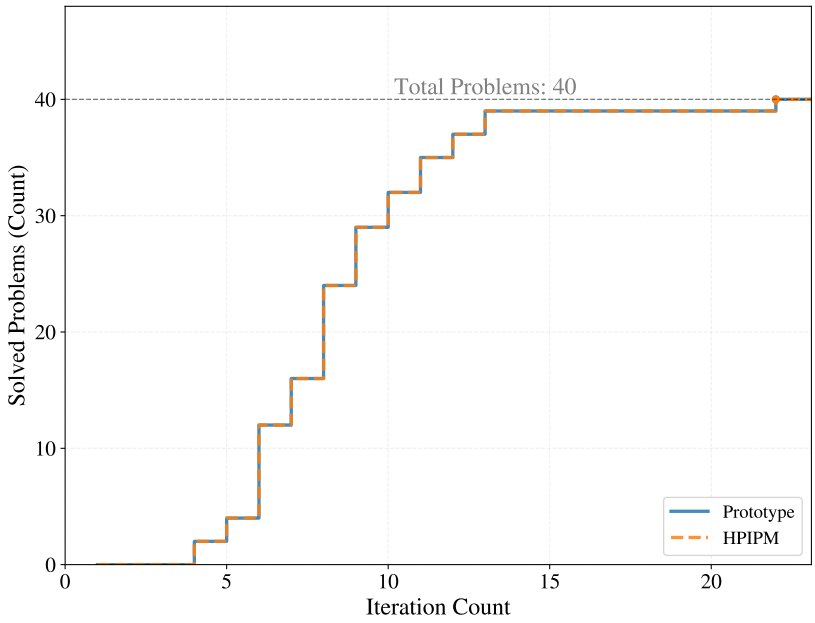


Figure 5.1: Iteration profiles on 40 convex QPs. The prototype solver is compared with HPIPM, showing identical iteration behavior.

5.3 Experiment setup

In this section, the settings for the experiment are introduced. First, based on the previously discussed methods, we will name and classify the regularization methods considered in the experiment, then we will introduce the general setup of the experiments, including the convergence criterion and step size control strategy.

5.3.1 Evaluated regularizations

Before turning to the results, we fix the set of regularization methods entering the comparison, together with the naming convention used throughout this chapter.

In the experiment, the methods are grouped by their *target*, that is, by the matrix which the correction is derived from, and the correction is directly applied to. This yields three categories: full Hessian regularization, Riccati-based regularization, and inertia correction. The numerical strategies of Section 3.1.1 are largely independent of the matrix they act on and can therefore be paired with each target. Every method is thus accordingly named in the form TARGET_STRATEGY, so that both factors can be read off directly from the label, and the comparison can be carried out either across targets at a fixed strategy or across strategies at a fixed target.

Full Hessian Regularization The first category follows Section 3.1.3 and takes the full Hessian as the target matrix. The regularization is triggered by the failure of the Cholesky factorization in Riccati recursion, and is applied stage-wise to each full Hessian block. The Riccati recursion is then re-attempted with the regularized full Hessian. Applying the three strategies of Section 3.1.1 to each block $\tilde{H}_k$ gives H_PROJECT, H_MIRROR and H_GMW.

Riccati-based Regularization The second category is the one proposed in Section 3.2 and acts on the reduced Hessian $\tilde{M}$. The algorithms are given in Algorithms 4 and 5, where the regularization is applied stage-wise to the block reduced Hessian $\tilde{M}_k$ during the backward pass, when the failure of the Cholesky factorization of $\tilde{M}_k$ occurs.

The two algorithms differ in which matrices of the backward pass the adjustment is derived from and applied to, giving two targets within this category:

- M: only the block reduced Hessian $\tilde{M}_k$ is corrected, as in Algorithm 4. The correction is purely local and serves as a minimal safeguard for the factorization.
- PM: the cost-to-go matrix P_{k+1} entering the stage is corrected as well, as

in Algorithm 5 introduced in this work, so that the propagation of negative curvature analyzed in Section 3.3 is accounted for consistently in $\tilde{M}_k$, Γ_k and P_k.

Pairing each target with the three strategies gives six methods, which test the two insights of Section 3.3, abandoning the minimum perturbation rule and correcting the cost-to-go matrix, separately and in combination. M_PROJECT applies neither and serves as the reference, M_MIRROR and M_GMW apply only the first, PM_PROJECT only the second, and PM_MIRROR and PM_GMW apply both.

Inertia Correction The third category is the LM-style inertia correction widely used in IPOPT [16], Fatrop, DDP [33] which we denote by INERTIA_LM.

In this strategy, a scalar multiple of the identity $\delta_w I$ is added to each full Hessian block $\tilde{H}_k$, i.e. $\tilde{H}_k + \delta_w I$, and the Riccati recursion is repeated with increasing δ_w until it succeeds. The trial sequence follows IPOPT: the unmodified matrix is tried first, then δ_w starts from δ_w^0, or from the last successful value reduced by κ_w^-, and is increased geometrically until the Riccati recursion succeeds or δ_w exceeds $\delta_w^{\max}$. The parameters are the IPOPT defaults listed in Table 5.2. Only the primal regularization from IPOPT is retained, as dual regularization is not the primary focus of this work. This method sits between the two preceding

Table 5.2: Inertia correction parameters.

Symbol	Value	Description
δ_w^0	10^{-4}	Initial Hessian perturbation parameter
$\delta_w^{\min}$	10^{-16}	Minimum allowable regularization parameter
$\delta_w^{\max}$	10^{16}	Maximum allowable regularization threshold
κ_w^-	$1/3$	Factor to decrease perturbation upon success
κ_w^+	8	Factor to increase perturbation after inertia failure
$\bar{\kappa}_w^+$	10^2	Accelerated scaling factor for severe indefiniteness
δ_{LM}	0.0	Initial LM shift (unmodified matrix attempted first)

categories, as it has no distinct target matrix in the above sense. Being a uniform multiple of the identity, the shift is not tied to any particular direction, so the perturbation remains a uniform shift along every direction for both the full and the reduced Hessian. Accordingly, as $\delta_w I$ is applied uniformly to Q_k and R_k in

full Hessian blocks $\tilde{H}_k$, it enters the Riccati recursion as the same additive term on the reduced Hessian blocks $\tilde{M}_k$ and P_k.

Taken together, the three categories span ten methods, which share the same solver, the same benchmark and the same parameters introduced in the next subsection.

5.3.2 General setup

All ten methods are run under the same conditions. The barrier parameter is updated by Mehrotra's probing and the step size is controlled by the FTB rule followed by the two-phase line search, while the remaining settings are collected in Table 5.3.

A run terminates either when all four residuals fall below their tolerances or when the iteration budget $i_{\max}$ is exhausted, in which case the instance counts as unsolved.

Table 5.3: Settings shared by all experiments.

Symbol	Value	Role
r_{stat}	10^{-6}	Tolerance on the stationarity residual
r_{eq}	10^{-8}	Tolerance on the equality residual
r_{ineq}	10^{-8}	Tolerance on the inequality residual
r_{comp}	10^{-6}	Tolerance on the complementarity residual
$i_{\max}$	120	Iteration budget, beyond which the instance counts as unsolved
$n_{\mathcal{P}}$	175	Nonconvex QP instances in the test set
ϵ	10^{-4}	Floor on the corrected eigenvalues and pivots, shared by all strategies
β	0.4	Backtracking factor, $\alpha_{\text{prim}} \leftarrow \beta\alpha_{\text{prim}}$
$n_{\max}$	10	Maximum backtracking iterations
κ_s	10^{-1}	Switching constant, $\alpha_{\text{prim}}(-\rho) > \kappa_s\,\theta^2$
γ	10^{-4}	Armijo constant for the f-phase, $\phi^+ \leq \phi_i + \gamma\,\alpha_{\text{prim}}\rho$

5.4 Numerical comparison

In this section, we report the numerical results of the ten regularization methods on the nonconvex QPs of the benchmark. The comparison proceeds in three stages, each narrowing the field. We first compare the six Riccati-based methods against one another, in order to identify the strongest representatives of the category. These are then set against the full Hessian regularizations, and finally against inertia correction, which serves as the practical reference point.

5.4.1 Comparison among Riccati-based regularizations

We first compare the proposed Riccati-based regularization methods, which are the main focus of this work. As a baseline we also include the unregularized recursion (NO_REGULARIZE).

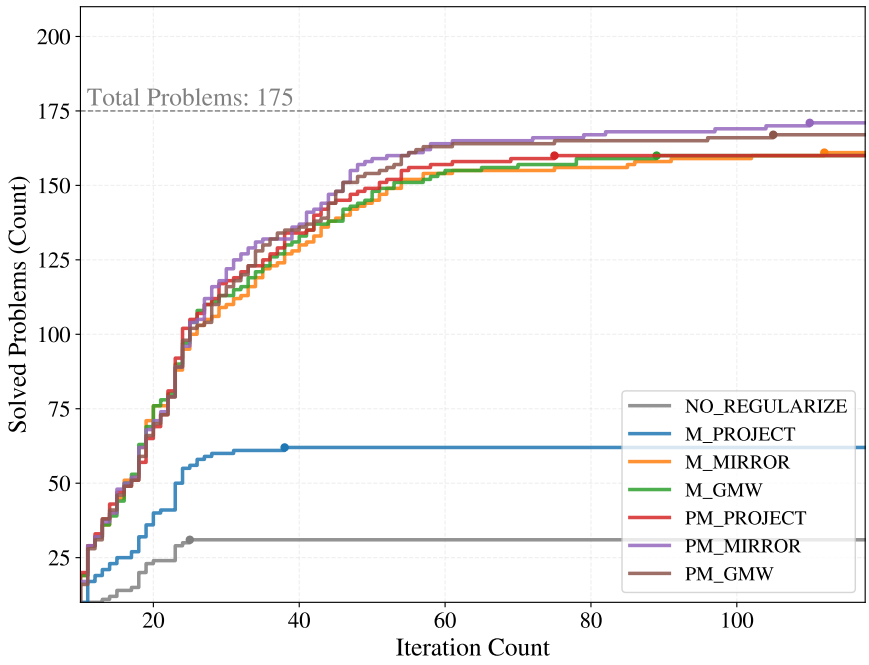


Figure 5.2: Iteration profile of the six Riccati-based methods over the 175-instance test set: number of instances solved within a given iteration budget.

The iteration profile in Figure 5.2 reports the number of instances solved within a given iteration budget for each method, where two trends stand out. First, each PM dominates its counterpart M once the budget exceeds about 30 iterations,

second, among the regularization strategies, MIRROR and GMW are clearly more effective than PROJECT. These trends validate the insights mentioned in Section 3.3. Among all the methods, PM_MIRROR tops the profile, solving 171 of the 175 instances, with PM_GMW close behind at 167.

Besides, it is interesting to note that the unregularized recursion can also solve a non-negligible fraction of the test set (31 of the 175 instances), which can be explained by the curvature contribution from the barrier term that helps to make the reduced Hessian PD in some instances.

Figure 5.3 complements this picture by showing the distribution of the worst-case decomposition cost in flop estimation, annotated with the maximum observed cost and the number of failures for each method. Since only the successful instances and maximal iteration instances are counted, the M_PROJECT will be excluded from the analysis, as it fails on 113 of the 175 instances.

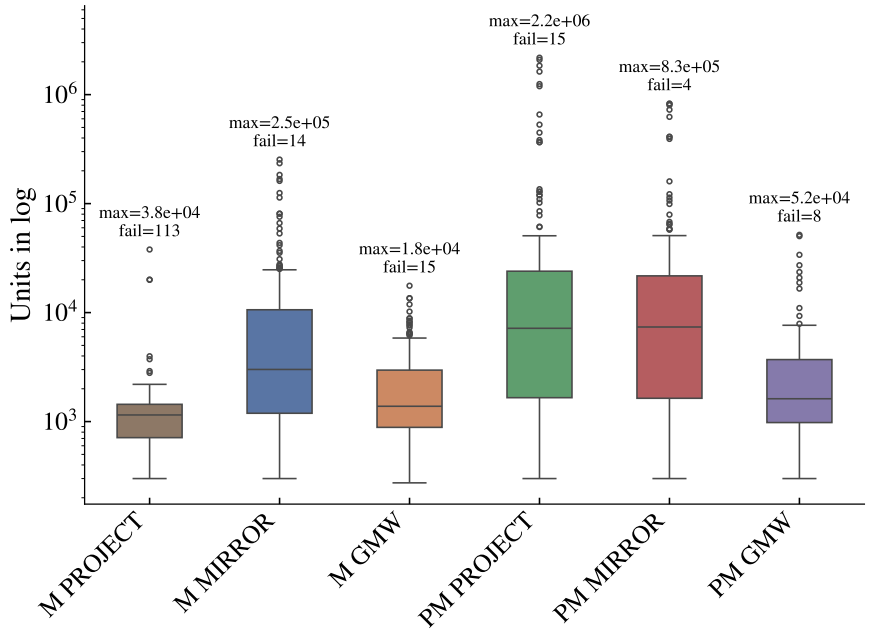


Figure 5.3: Worst-case decomposition cost of the six Riccati-based methods (flops, log scale). Annotations report the maximum observed cost and the number of unsolved instances.

In general, the PM methods incur a higher and more dispersed cost, reflecting the extra factorization work introduced by the cost-to-go term, yet they are generally more robust and thus the focus of the comparison. In Figure 5.3, PM_MIRROR is the most robust method with only 4 unsolved problems, and PM_GMW follows

with 8 while keeping its worst-case cost well below that of both PM_MIRROR and PM_PROJECT. Notably, though the PM_PROJECT and PM_MIRROR belong to the same class of regularization strategy, the worst-case cost of PM_PROJECT is higher than that of PM_MIRROR, which can be explained by the issue of inadequate regularization in the minimum perturbation strategy, so regularization is triggered more often in the PM_PROJECT case.

Taken together, the two figures first reveal a robustness ordering, PM_MIRROR $>$ PM_GMW $>$ {M_MIRROR, M_GMW, PM_PROJECT} $>$ M_PROJECT, where PM_PROJECT only matches the plain M methods. If reliability is the priority, PM_MIRROR is the safest choice, as it leaves the fewest instances unsolved. If a balance between reliability and computational cost is preferred, PM_GMW is the more attractive option, which maintains the robustness (8 unsolved instances) at a much lower worst-case cost. We will therefore focus on PM_MIRROR and PM_GMW in the following comparisons with other regularization categories.

5.4.2 Comparison with full Hessian regularizations

Having identified PM_MIRROR and PM_GMW as the strongest Riccati-based methods, we now compare them against the full Hessian regularizations.

The iteration profile in Figure 5.4 shows the two Riccati-based methods sitting well above every full Hessian variant across the whole iteration budget. PM_MIRROR and PM_GMW leave only 4 and 8 instances unsolved, whereas each H method leaves a substantial number of instances unsolved even at the largest budget. This suggests that regularizing the full Hessian does not recover the reliability or efficiency obtained by regularizing within the Riccati recursion.

The gap is even more pronounced in Figure 5.5, in both robustness and cost. The full Hessian methods fail on 28 (H_PROJECT), 40 (H_MIRROR) and 57 (H_GMW) instances, against 4 for PM_MIRROR and 8 for PM_GMW. The worst-case costs of H_PROJECT and H_MIRROR lie far above those of both Riccati-based methods, and even H_GMW, the cheapest and least robust full Hessian method, has a higher median cost than PM_GMW.

The Riccati-based methods are simultaneously more robust and cheaper than

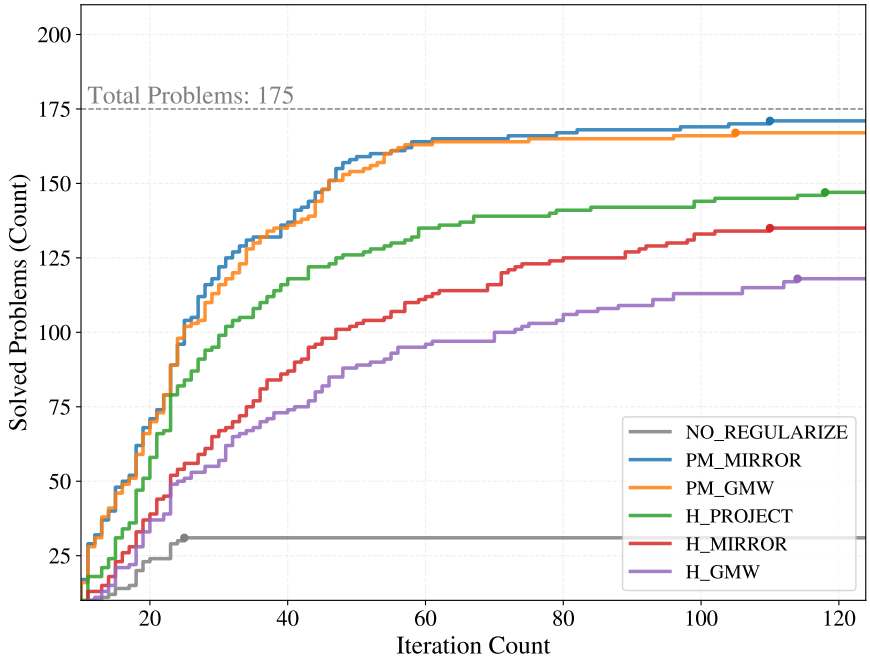


Figure 5.4: Iteration profile of the two strongest Riccati-based methods (PM_MIRROR, PM_GMW) against the full Hessian regularizations (H_PROJECT, H_MIRROR, H_GMW).

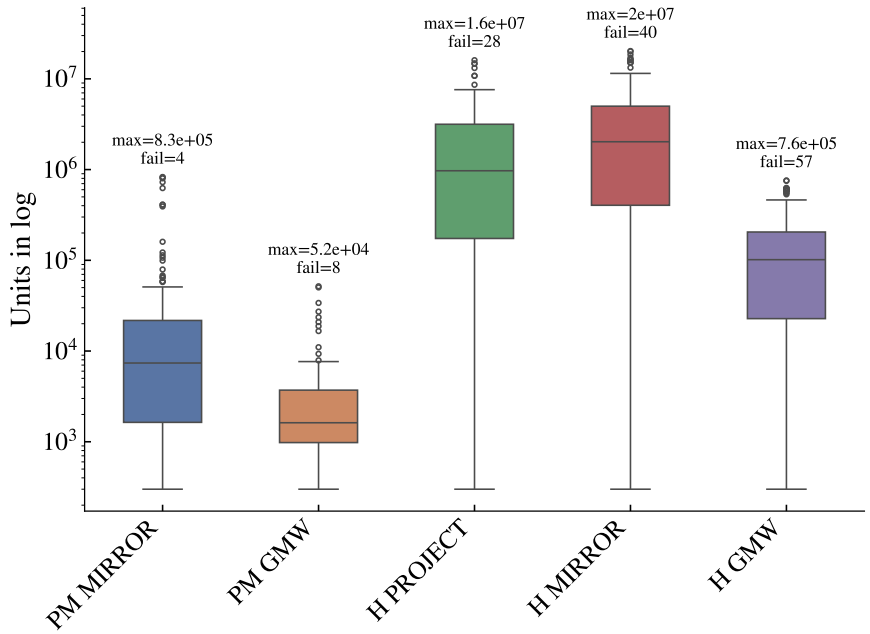


Figure 5.5: Worst-case decomposition cost of the two strongest Riccati-based methods against the full Hessian regularizations (flops, log scale). Annotations as in Figure 5.3.

every full Hessian counterpart in Figure 5.5. This generally supports regularizing the reduced Hessian by exploiting the stage-wise structure of the problem rather than the full Hessian.

5.4.3 Comparison with inertia correction

Finally, we compare the recommended Riccati-based methods against inertia correction (INERTIA_LM), which is the commonly used strategy in practice.

The iteration profiles in Figure 5.6 end close for the three regularized methods, with PM_MIRROR and PM_GMW solving the most instances (171 and 167) and INERTIA_LM close behind (164). However, beyond about 25 iterations the PM_MIRROR and PM_GMW sit clearly above INERTIA_LM, indicating that fewer iterations are needed to solve the same number of instances.

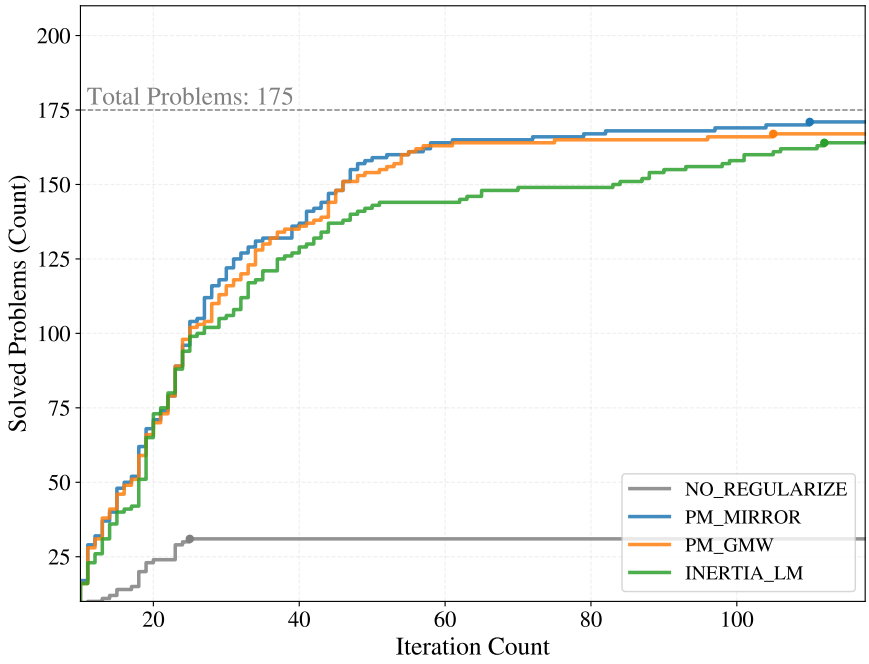


Figure 5.6: Iteration profile of PM_MIRROR and PM_GMW against inertia correction (INERTIA_LM).

The decomposition-cost distribution in Figure 5.7 draws the distinction more finely. Inertia correction is the cheapest of the three in the worst case, with a maximum cost of 2.5×10^4 flops, below both PM_GMW (5.2×10^4) and PM_MIRROR (8.3×10^5). In terms of median cost, however, INERTIA_LM and PM_GMW are

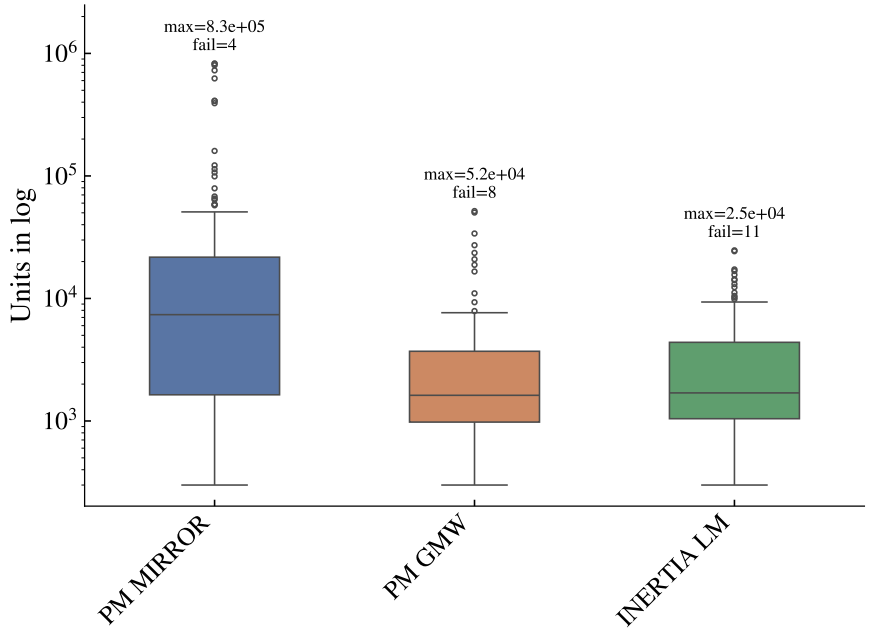


Figure 5.7: Worst-case decomposition cost of PM_MIRROR and PM_GMW against inertia correction (flops, log scale). Annotations as in Figure 5.3.

essentially on par, with PM_GMW marginally lower. Their robustness differs, however: INERTIA_LM fails on 11 instances, against 8 for PM_GMW and 4 for PM_MIRROR.

From this benchmark, inertia correction is therefore preferable only where a hard bound on the per-decomposition cost outweighs the number of solved instances.

5.4.4 Ablation on the conditional Mehrotra

All results so far use Mehrotra's probing. To justify this choice, we rerun PM_MIRROR, PM_GMW, INERTIA_LM and H_PROJECT, together with the unregularized baseline, using the conditional predictor–corrector instead, with all other settings unchanged.

Figure 5.8 shows that every method solves fewer instances with the corrector, but the loss is far from uniform. PM_MIRROR and PM_GMW drop from 171 and 167 to 91 and 85 solved instances, whereas INERTIA_LM and H_PROJECT only drop from 164 to 149 and from 147 to 138. The ranking of Section 5.4 is thereby

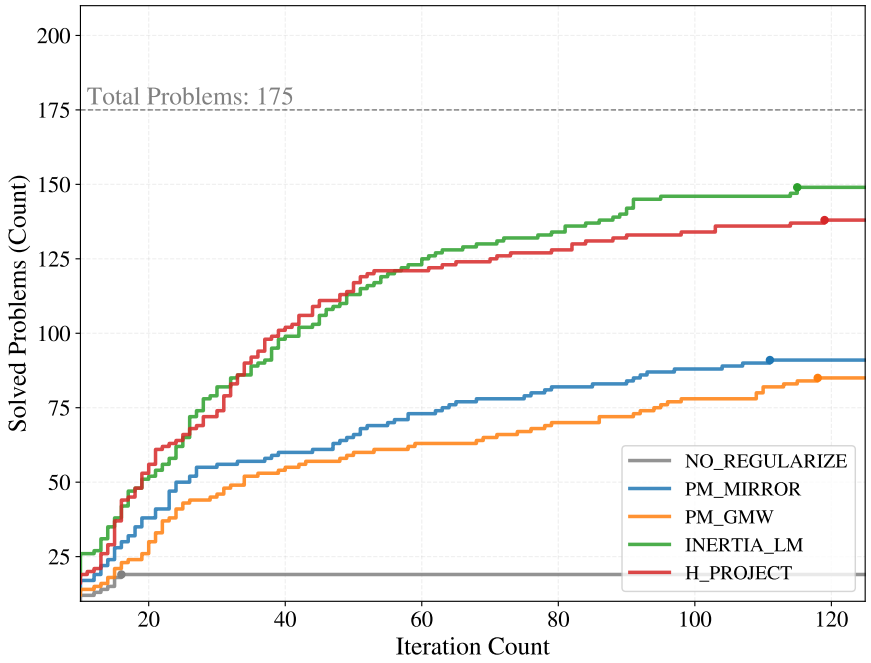


Figure 5.8: Iteration profile with the conditional predictor–corrector of PM_MIRROR, PM_GMW, INERTIA_LM, H_PROJECT and the unregularized baseline.

reversed, and the two Riccati-based methods now trail both alternatives.

This is consistent with the argument in Section 2.3 and supports probing as the default method for nonconvex QPs, as it improves the performance of methods generally. Section 5.5.3 discusses why PM_MIRROR loses far more instances than INERTIA_LM.

5.4.5 Ablation on the line search

The two-phase line search of Section 2.4.2 was introduced to safeguard the FTB step on nonconvex QPs. To isolate its contribution, we disable it, so that the FTB step is taken directly, and rerun the same five methods as in the previous subsection with all other settings unchanged.

Figure 5.9 shows that the line search has a much smaller effect than the choice of barrier update. Without it, PM_MIRROR solves 167 instead of 171 instances, PM_GMW 166 instead of 167 and H_PROJECT 146 instead of 147, while IN-

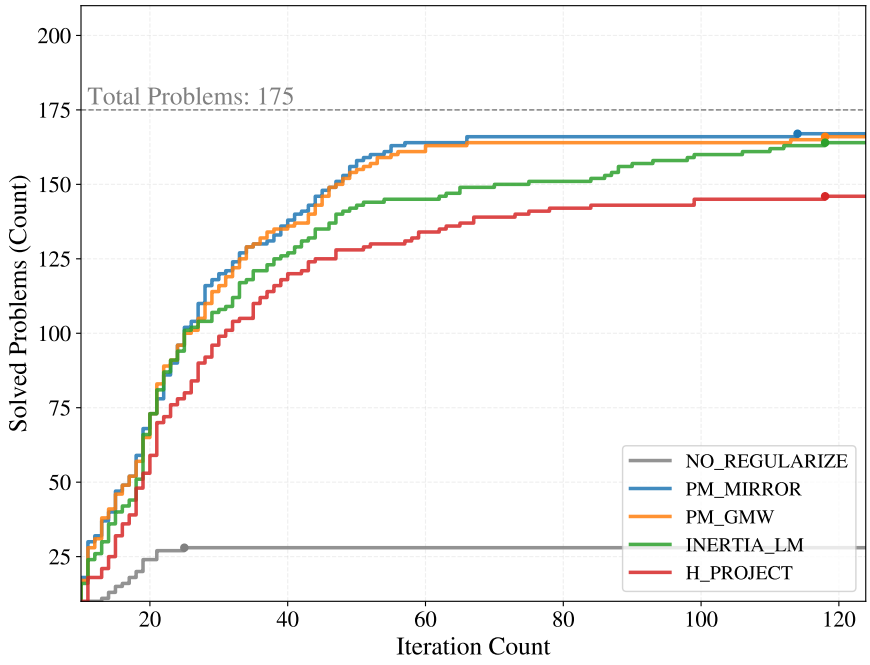


Figure 5.9: Iteration profile without the line search of PM_MIRROR, PM_GMW, INERTIA_LM, H_PROJECT and the unregularized baseline.

ERTIA_LM solves 164 in both cases. PM_MIRROR benefits most, as the line search adds four solved instances and loses none, whereas for the other methods the instances it gains and loses nearly cancel. On the instances solved in both settings, most iteration counts are identical.

The ranking of Section 5.4 therefore does not rely on the line search. The robustness of the Riccati-based methods stems from the regularization itself.

5.5 Discussion

This section first resolves the failures in experiments along two directions. Section 5.5.1 splits them by the status with which the solve terminated, which separates a numerical breakdown of the regularized recursion from a mere exhaustion of the iteration budget. Section 5.5.2 splits them by the OCP each instance was collected from, which identifies what distinguishes the hard instances from the easy ones and explains why some methods perform poorly. Finally, Section 5.5.3 explains why the conditional Mehrotra corrector affects PM_MIRROR

and INERTIA_LM so differently.

5.5.1 Failure by status

A run that does not converge ends with one of three exit statuses, shown per method in Figure 5.10:

- *Unregularized*: the Cholesky factorization fails *again* after the regularization has been applied, usually caused by numerical indefiniteness.
- *Max iterations reached*: the residuals did not fall below their tolerances within the budget $i_{\max} = 120$.
- *NaN encountered*: an infinite or undefined value appeared in the residuals, so that the iterate itself has overflowed.

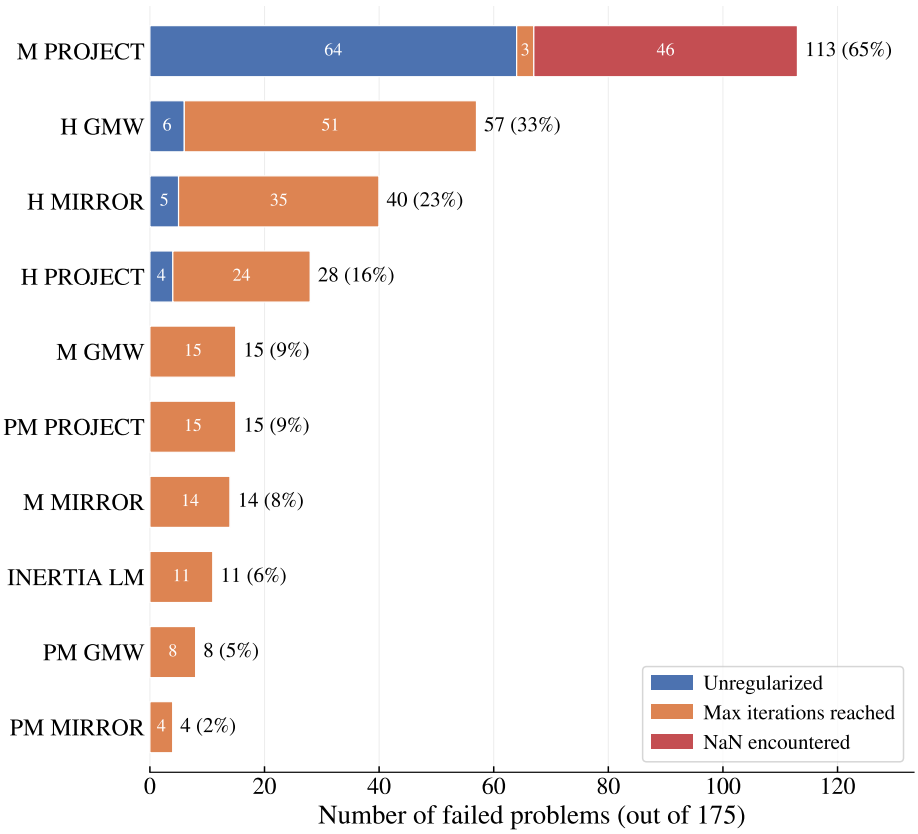


Figure 5.10: Failures of each method by exit status over the 175-instance test set. The bars are stacked by status and annotated with the total number of unsolved instances and the corresponding share of the test set.

In Figure 5.10, the M_PROJECT is the only Riccati-based method with numerical breakdowns, with 113 of its 175 instances unsolved. 110 of its 113 failures stem from numerical issues, while only 3 exhaust the iteration budget. This confirms the analysis in Section 3.3, where ill-conditioning propagates into numerical indefiniteness or overflow. Adopting either remedy identified there resolves these breakdowns: abandoning the minimum perturbation rule (M_MIRROR, M_GMW) or adding the cost-to-go correction (PM_PROJECT) eliminates all NaN and unregularized failures, leaving only iteration-limit terminations (14, 15, and 15 failures, respectively). Combining both remedies, as in PM_MIRROR and PM_GMW, further reduces the remaining iteration-limit failures.

The full Hessian methods still break down on 4, 5 and 6 instances, but most of their failures reach the iteration limit (24, 35 and 51). Their breakdowns come from the numerical indefiniteness that arises in the IPM.

In the nonconvex case, local minimizers frequently lie on the constraint boundary. Near convergence, the barrier term $\Lambda_k T_k^{-1}$ grows large as the slacks of active constraints approach zero, which enters $\tilde{M}_k$ both directly through $\tilde{R}_k$ and recursively through P_{k+1}. Once $\kappa_2(\tilde{M}_k)\,\epsilon_{\text{mach}} \geq 1$, where ϵ_{mach} denotes the machine precision, rounding errors exceed the smallest eigenvalue of $\tilde{M}_k$, causing its Cholesky factorization to fail.

For the full Hessian methods, although every $\tilde{H}_k^{\text{reg}}$ is already PD, the Cholesky factorization of $\tilde{M}_{k-1}$ could still fail due to rounding errors. The Riccati-based methods, however, correct $\tilde{M}_{k-1}$ itself whenever its factorization fails, so the indefiniteness is handled whether it stems from the problem or from rounding.

5.5.2 Failure by category

Figure 5.11 resolves the same failures by the OCP each instance was collected from. Combined with Figure 4.1, it suggests that an instance becomes hard only when its indefiniteness is both deep and wide. Deep but narrow indefiniteness, as in *Crane* and *Flying robot*, and wide but moderate indefiniteness, as in *Dubins car* and *Robot arm*, both remain solvable, as PM_MIRROR solves every instance of these four OCPs. *Acrobot* is the only OCP that combines both, and its four unsolved instances defeat every method within the iteration budget.

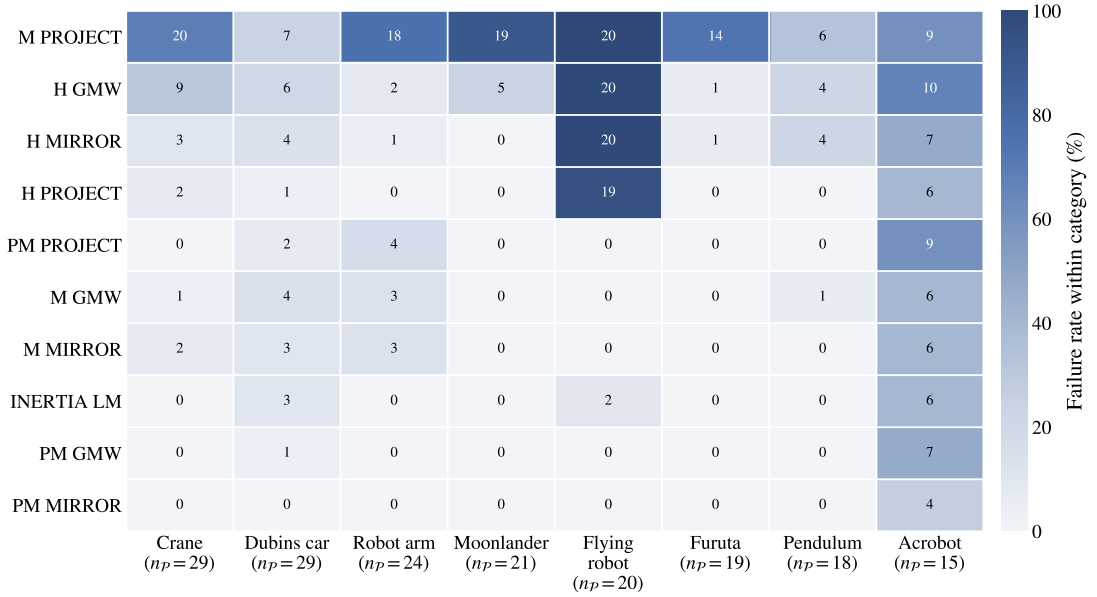


Figure 5.11: Failures of each method by OCP category. The cell entries give the number of unsolved instances, and $n_{\mathcal{P}}$ is the number of instances collected from that OCP. The rows are ordered by total failure count.

For the full Hessian methods, *Flying robot* is the hardest OCP, where H_PROJECT, H_MIRROR and H_GMW fail on 19, 20 and 20 of its 20 instances, while every Riccati-based method except M_PROJECT solves all of them. Recalling Figure 4.2, in *Flying robot*, the indefiniteness lies only in the cost-to-go matrix P_k and the indefinite-stage rate is low. The full Hessian methods cannot act on P_k, so they perturb $\tilde{Q}_k$ over the whole horizon, including the stages that need no correction, and exhaust the iteration budget. This reveals an advantage of the Riccati-based regularization, which treats indefiniteness where it arises in the recursion.

INERTIA_LM adds $\delta_w I$ to both $\tilde{Q}_k$ and $\tilde{R}_k$, which in the Riccati recursion shifts both P_k and $\tilde{M}_k$. It therefore solves 18 of the 20 *Flying robot* instances, and its total failure count is much closer to the PM methods than to the full Hessian ones. Unlike the Riccati-based methods, however, its shift is not localized: δ_w must be large enough for the most negative eigenvalue over all stages and is applied at every stage.

5.5.3 Analysis of the Mehrotra ablation

Based on the ablation in Section 5.4.4, this subsection explains why the ranking of Section 5.4 reverses under the conditional Mehrotra corrector, i.e. why PM_MIRROR loses far more instances than INERTIA_LM.

Recall Equation (2.9). With simplified notation, the regularized system reads

$$(\tilde{M}_i + E_i)\Delta y_z = -r_i, \tag{5.6}$$

and the step is bounded by

$$\|\Delta y_z\| \leq \frac{\|r_i\|}{\nu_{\min}(\tilde{M}_i + E_i)}. \tag{5.7}$$

As one can notice, the regularization would shorten the step taken, and the methods differ in how small $\nu_{\min}(\tilde{M}_i + E_i)$ can become, and thus in how long the regularized step is.

INERTIA_LM increases a uniform shift δ until the factorization succeeds. If $\delta_{\min}$ is the smallest shift for which it succeeds, then $\nu_{\min}(\tilde{M}_i + E_i) \geq \delta - \delta_{\min}$. Since the shift is increased by the factor κ_w^+, the accepted δ usually lies well above $\delta_{\min}$, and $\nu_{\min}$ stays of the order of δ.

PM_MIRROR modifies only the negative eigenvalues of $\tilde{M}_k$ and P_k, stage by stage. A small positive pivot passes the factorization unmodified, so nothing prevents $\nu_{\min}(\tilde{M}_i + E_i)$ from approaching zero. PM_MIRROR could reach a much smaller $\nu_{\min}(\tilde{M}_i + E_i)$ than INERTIA_LM at similar $\|r_i\|$, and its steps are correspondingly longer than those of INERTIA_LM.

Mehrotra's corrector adds the second-order term $\Delta T_{\text{aff}} \Delta \Lambda_{\text{aff}} \mathbf{1}$ to the RHS, which grows quadratically with the affine step. As shown in Table 5.4, the median affine step of PM_MIRROR is far longer than that of INERTIA_LM.

In PM_MIRROR, the long affine step makes the second-order term dominate the RHS, so the corrector step is long as well. For both steps, the FTB rule then allows only a short step size, which keeps μ_{aff} and μ_{cor} close to μ, so μ_{cor} remains below $2\mu_{\text{aff}}$. The safeguard of Section 2.3 rarely detects this, so the bad corrector step is taken, which is dominated by the second-order error of an affine step, so

it neither reduces the residuals nor moves the iterate along the directions needed to leave the indefinite region.

INERTIA_LM, however, keeps the second-order term comparable in size to the rest of the RHS, so the corrected direction remains moderate as well, and the FTB rule leaves a step size comparable to that of a centering step. Since μ_{aff} then lies clearly below μ, the safeguard is effective and rejects 39.1% of the corrector steps. The conditional corrector step thus makes progress as intended, and INERTIA_LM loses far fewer instances to the corrector than PM_MIRROR.

Table 5.4: Step metrics of PM_MIRROR and INERTIA_LM.

	PM_MIRROR	INERTIA_LM
Corrector, on 175 *instances*		
Median affine step $\|\Delta y_z\|$	65.4	4.4
Percentage of rejected corrector steps	3.7 %	39.1 %
Probing, on 164 *instances solved by both*		
Iterations	4218	5132
Regularized iterations	1566	2289
Median displacement $\alpha_{\text{prim}}\|\Delta y_z\|$	1.44	0.47

With probing, the affine step only determines the centering parameter, so the second-order term no longer enters the search direction. The longer step of PM_MIRROR, which is a drawback under the corrector, thus becomes an advantage, as measured by the displacement $\alpha_{\text{prim}}\|\Delta y_z\|$ in Table 5.4. INERTIA_LM moves less per regularized iteration and needs more of them, while PM_MIRROR moves further per regularized iteration. This larger displacement could help the iterate to move towards the constraint boundaries, where the reduced Hessian could be PD with curvature contribution, and recover the fast convergence towards solutions.

Chapter 6

Conclusion and outlook

This thesis regularized the indefinite QPs of exact-Hessian SQP for NMPC inside the interior point solver, where the barrier curvature is already included and only the reduced Hessian has to be positive definite. Since the Riccati backward pass is a block $\mathrm{LDL}^\top$ factorization of the reduced Hessian, indefiniteness is detected and corrected stage-wise wherever the Cholesky factorization of $\tilde{M}_k$ fails, and the recursion stays linear in the horizon length.

We showed that a minimum perturbation correction of $\tilde{M}_k$ adds a negative rank-one term to the cost-to-go matrix, which can make the preceding stages indefinite and ill-conditioned. The improved strategy therefore uses a non-minimal correction and also regularizes the cost-to-go matrix. We proved that the regularization becomes inactive close to the solution, so the local convergence rate of the interior point method is retained.

In the benchmark of 175 indefinite QPs from eight OCPs, the cost-to-go matrix is indefinite in most indefinite stages. With Mehrotra's probing, PM_MIRROR left 4 instances unsolved and PM_GMW 8 at a lower worst-case decomposition cost than PM_MIRROR, against 28 to 57 for full Hessian regularization and 11 for inertia correction. The full Hessian methods failed on almost every *Flying robot* instance, where only the cost-to-go matrix is indefinite, and only at a few stages. The instances that no method solved combine deep negative curvature with a large share of indefinite stages.

Several questions remain open:

- **Regularization magnitude.** The eigenvalue floor ϵ is fixed, and the remaining failures occur where large corrections are applied at many stages. Adapting the magnitude, e.g. to the conditioning of the corrected blocks or to the barrier parameter, could improve robustness.
- **Other MPC formulations.** In time-optimal and economic MPC, the cost provides little positive curvature, so indefiniteness is likely to be broader than in this benchmark. Whether the ranking of the methods carries over remains to be tested.
- **Embedding in SQP.** The QPs were solved in isolation with a Python prototype. An implementation in HPIPM and `acados` is needed to compare against SQP-level regularization in terms of SQP iterations, run time and closed-loop performance.

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